Data Converter Performance Metrics

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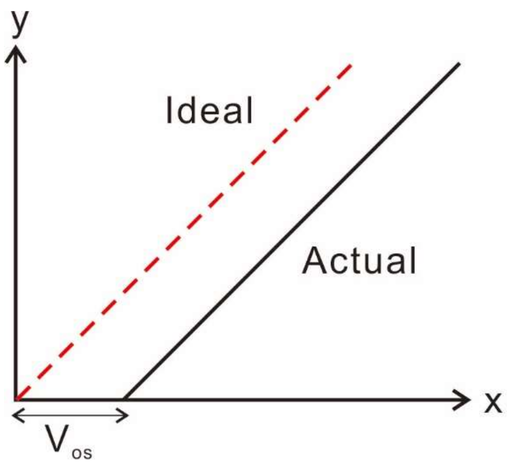
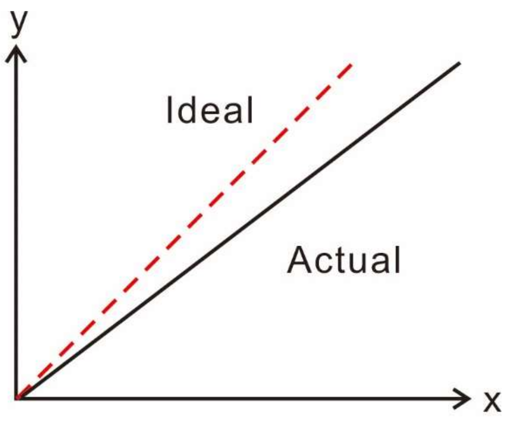
Static Performance Metrics

- Offset
- Gain error
- Differential nonlinearity (DNL)
- Integral nonlinearity (INL)

Basic Concept

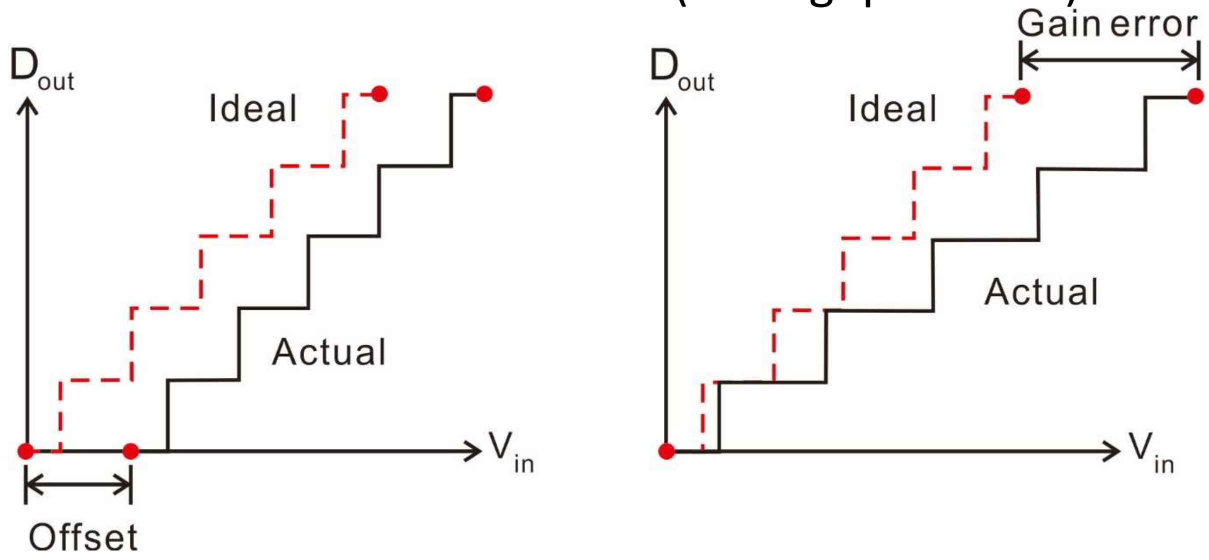




$$
\begin{aligned}
y &= ax + b \\
&= x + \underbrace{(a-1)}_{\text{Gain error}}x + \underbrace{b}_{\text{Offset error}}
\end{aligned}
$$

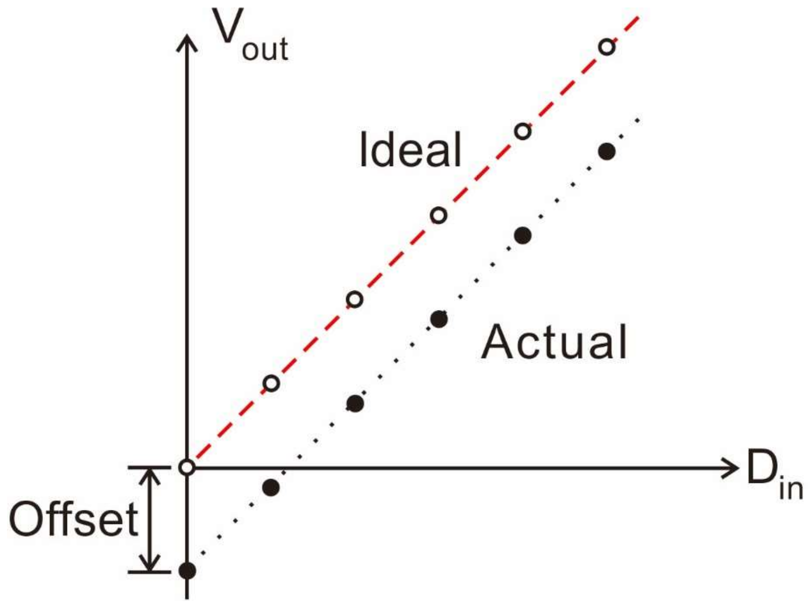
ADC Offset and Gain Error

- Start point: ½ LSB before first transition
- End point: ½ LSB after last transition
- **Offset**: the deviation of start point from its ideal location
- **Gain error**: the deviation of end point with offset removed
- Both quantities are measured in LSBs (analog quantities)



DAC Offset and Gain Error

- Start and end points are defined by analog output values corresponding to the smallest and largest digital inputs.
- Offset and gain errors are also measured in LSBs (analog quantities)



Comments

- Definitions in the previous slides are widely used in industry
  - Note that different companies may use different definitions
    - Read datasheet carefully
  - IEEE standard defines them based on least square curve fitting

- Gain and offset errors matters when we care about absolute magnitude
  - E.g., in scientific measurements

- However, we may not care about offsets and gain errors in many applications
  - These errors are easy to correct in the digital domain
  - Offsets and gain errors do not affect linearity
  - E.g., audio application, communication, etc

- We are more interested in nonlinearities.

Differential Nonlinearity (DNL)

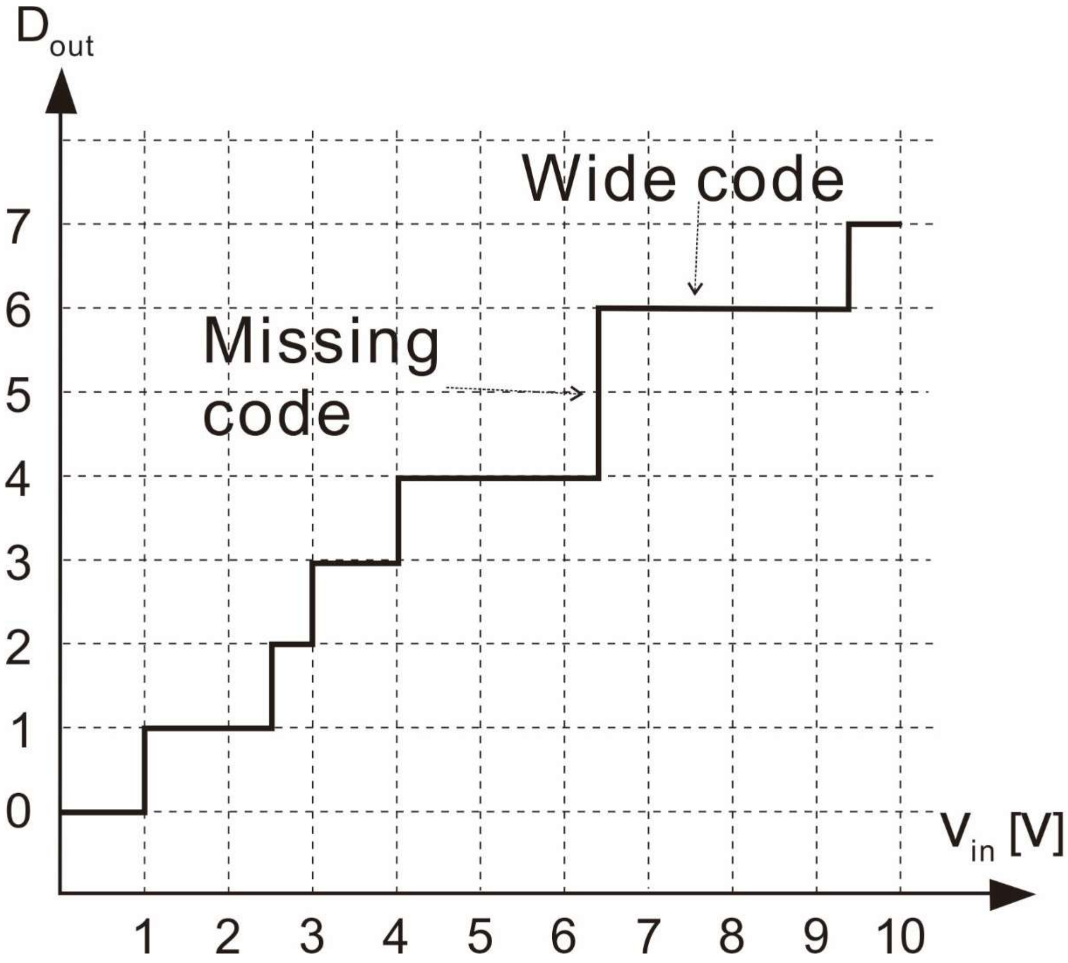
- For an ideal ADC, all codes have the same width.
- For an ideal DAC, all steps have the same size.
- However, in practice, the code widths and step sizes may not be identical.
- DNL[k] is defined as the deviation of the width from the "average" width for code k.
  - It is a measure of uniformity.
  - Gain and offset errors do not affect uniformity.

$$\mathrm{DNL[k]} = \frac{\mathrm{W[k]} - \mathrm{W_{avg}}}{\mathrm{W_{avg}}}$$

$$\mathrm{W[k]} = \mathrm{T[k+1]} - \mathrm{T[k]} \qquad \mathrm{W_{avg}} = \frac{\sum_{k=1}^{N} \mathrm{W[k]}}{\mathrm{N}} = \frac{\mathrm{T[N+1]} - \mathrm{T[1]}}{\mathrm{N}}$$

ADC DNL Example

Code [k]	T[k]	W [V]
0	undefined	undefined
1	1.0	1.5
2	2.5	0.5
3	3	1
4	4	2.4
5	6.4	0
6	6.4	3
7	9.4	undefined



ADC DNL Example

- What is the average code width?
  - There are in total 6 codes (not counting the first and last codes)
  - The total code width is 8.4V

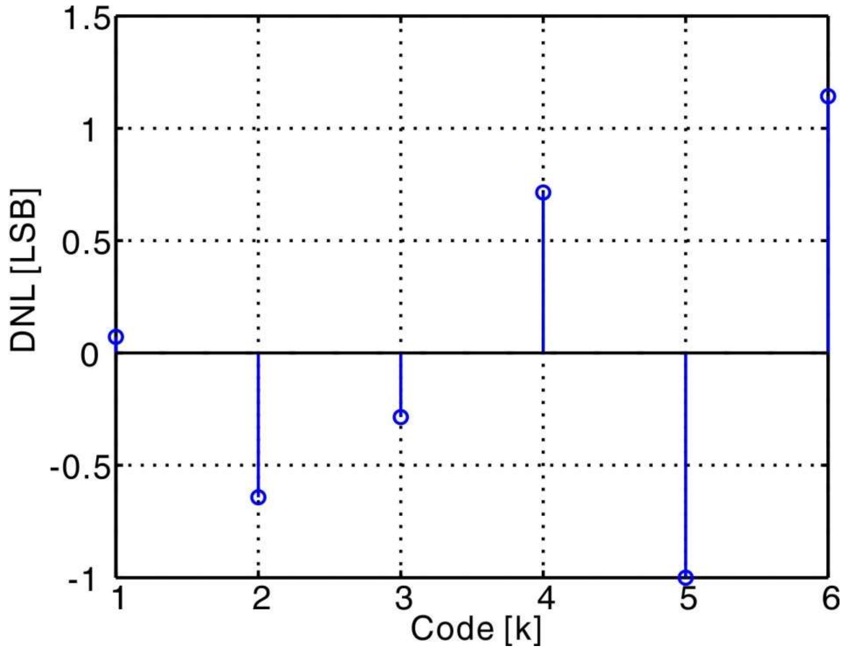
$$\mathrm{W_{avg}} = \frac{9.4\mathrm{V} - 1.0\mathrm{V}}{6} = 1.4\mathrm{V}$$

- DNL[k] is defined as

$$\mathrm{DNL[k]} = \frac{\mathrm{W[k]} - \mathrm{W_{avg}}}{\mathrm{W_{avg}}} = \frac{\mathrm{W[k]} - 1.4}{1.4}$$

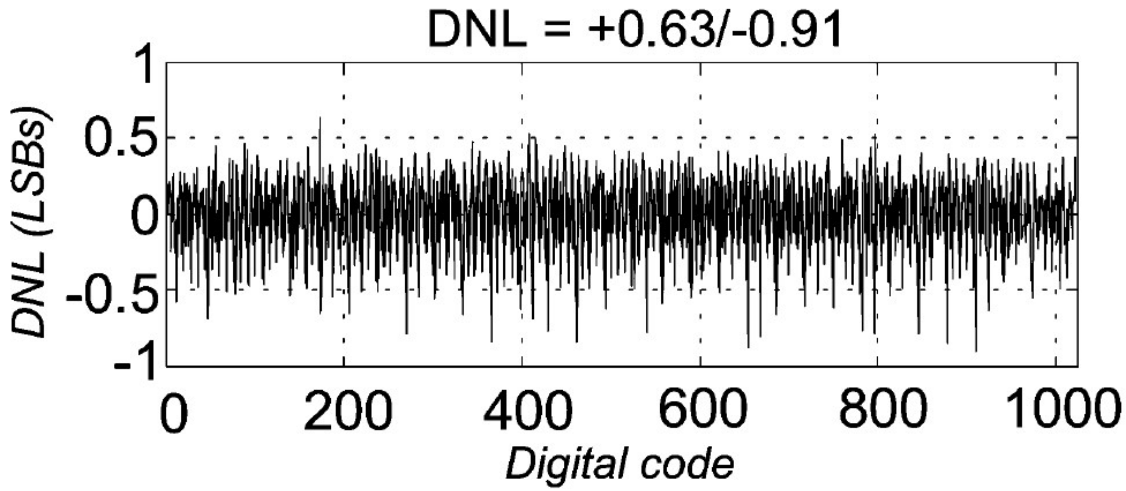
Result

Code [k]	DNL [LSB]
1	+0.07
2	-0.64
3	-0.29
4	+0.71
5	-1.00
6	+1.14



- Positive/negative DNL implies wide/narrow code
- DNL = -1 LSB implies missing code
- Possible to have DNL > +1 LSB
- Impossible to have DNL < -1 LSB for an ADC (why?)
- The sum of all DNL[k] is equal to zero (why?)

Typical ADC DNL Plot



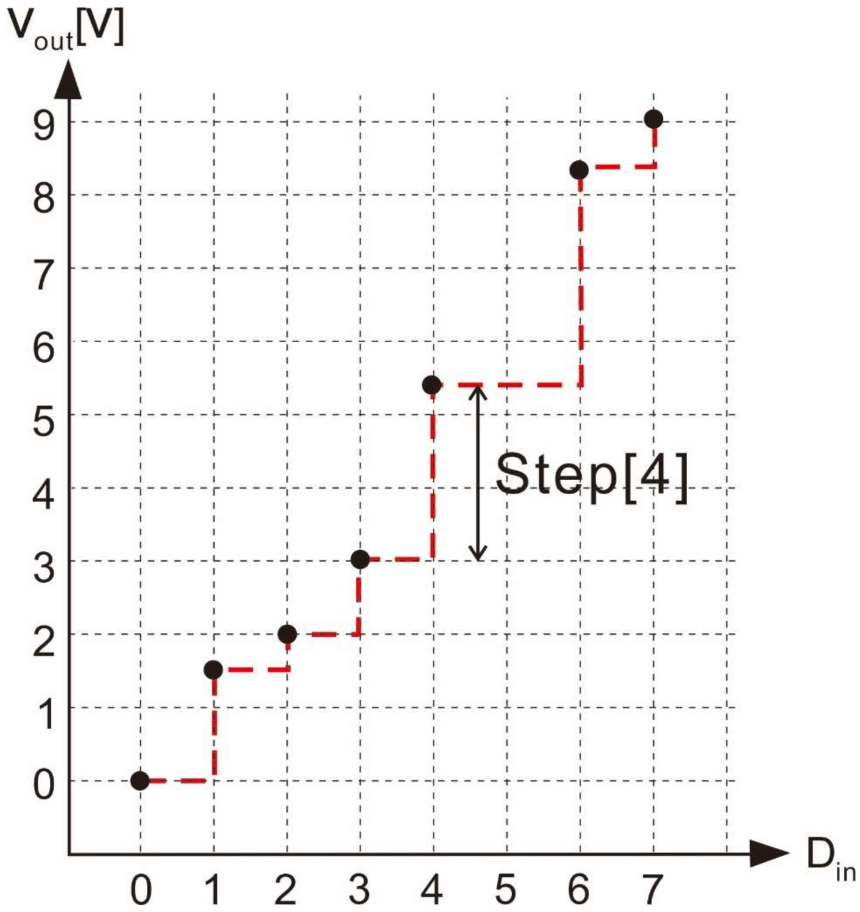
[Ahmed, JSSC 12/2005]

- Usually, we only specify the min/max DNL numbers across all codes
  - E.g. DNL = +0.63/-0.91 LSB
- A well designed ADC usually has DNL bounded by $\pm$1 LSB
- DNL of an ADC is usually obtained by using a linear ramp input and measuring the duration of all codes.

DAC DNL

- Same idea as that for an ADC.
  - Note that static metrics are all defined as analog values.

$$\mathrm{DNL[k]} = \frac{\mathrm{Step[k]} - \mathrm{Step_{avg}}}{\mathrm{Step_{avg}}}$$



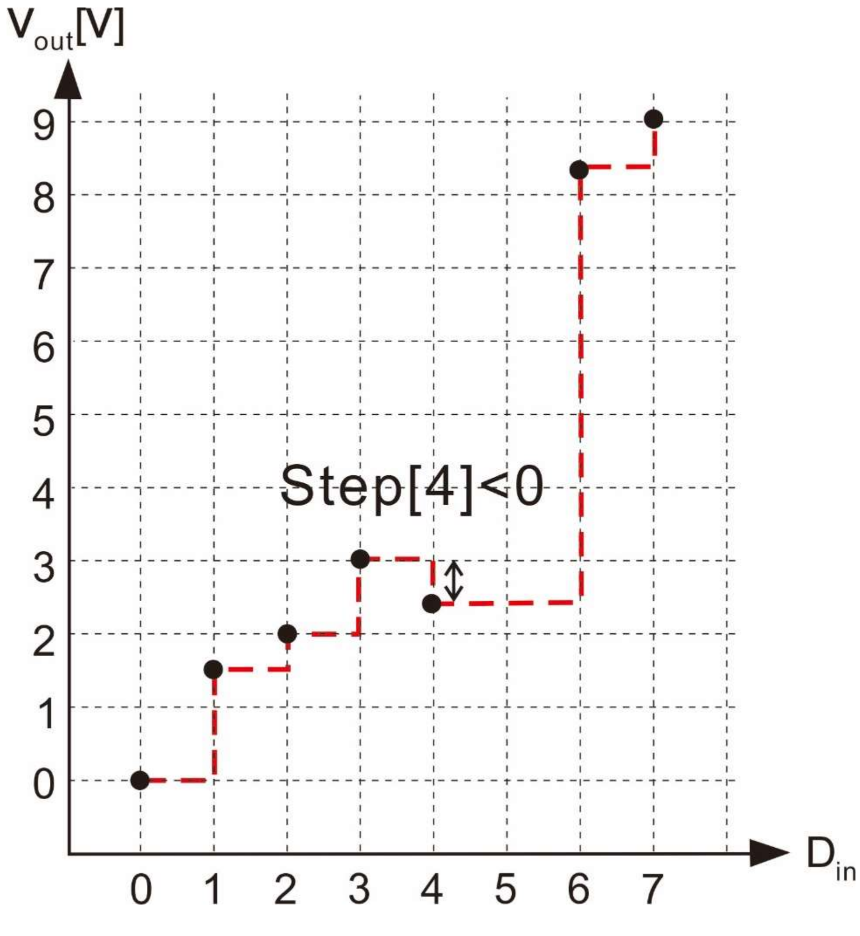
- DAC DNL can be less than -1 LSB. Why?

Non-Monotonic DAC

- DNL < –1 LSB implies non-monotonicity

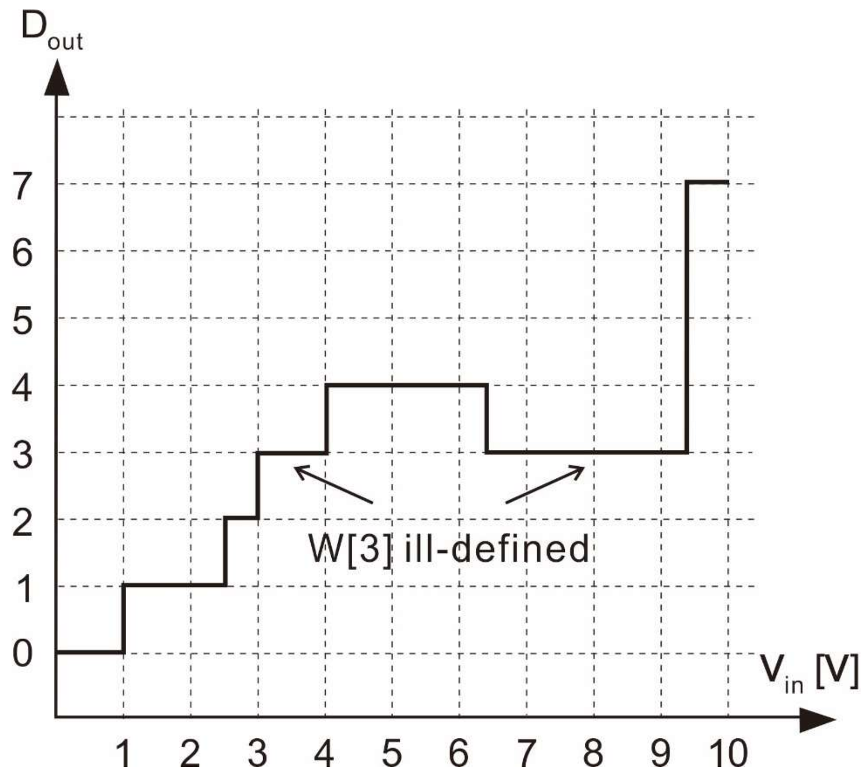
$$\mathrm{DNL}[4] = \frac{\mathrm{Step}[4] - \mathrm{Step}_{\mathrm{avg}}}{\mathrm{Step}_{\mathrm{avg}}} < -1 \ \ \mathrm{LSB}$$

- How about a non-monotonic ADC?



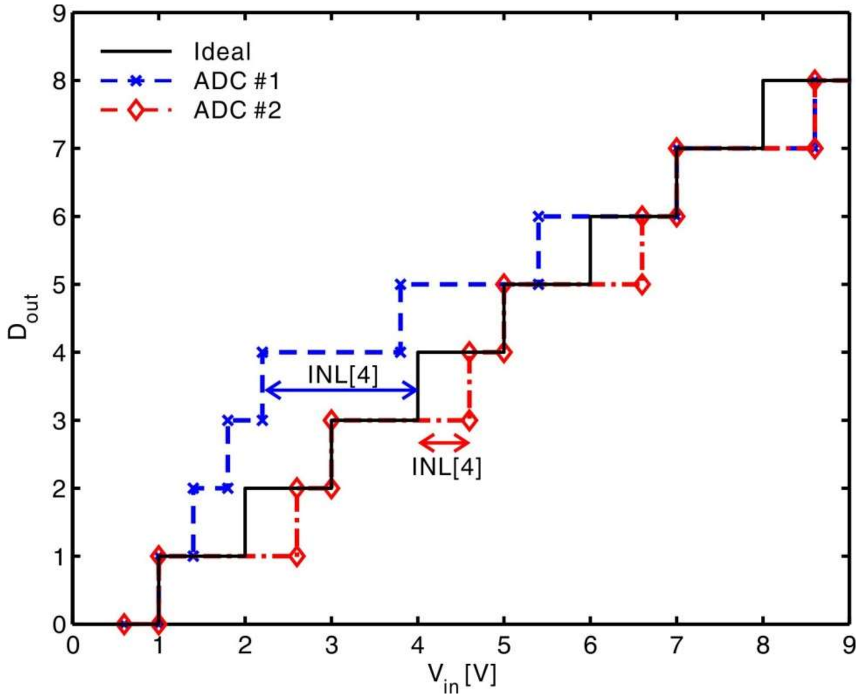
Non-Monotonic ADC

- Code 3 has 2 transition levels
- W[3] is ill-defined, and thus, DNL[3] is ill-defined.
- Not a major issue, because such a non-monotonic ADC will not be sold anyway.



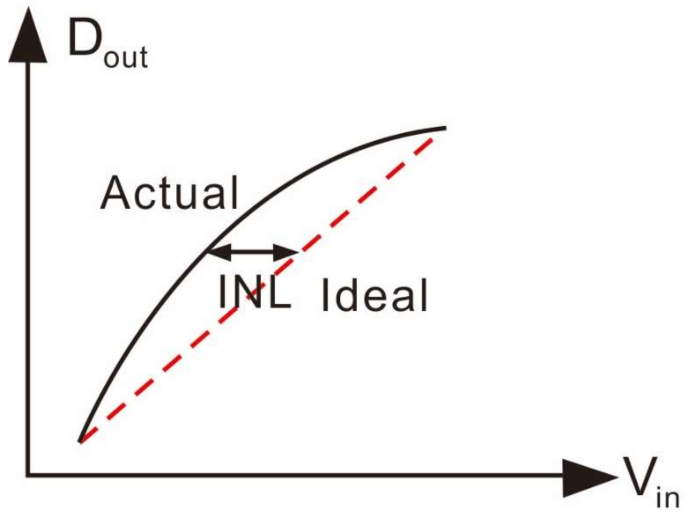
Integral Nonlinearity (INL)

- DNL, by itself, is insufficient for characterizing the nonlinearities.
- Compare the following two ADC, which one is more nonlinear?
  - ADC #1 DNL = [-0.6, -0.6, -0.6, -0.6, +0.6, +0.6, +0.6, +0.6]
  - ADC #2 DNL = [-0.6, +0.6, -0.6, +0.6, -0.6, +0.6, -0.6, +0.6]
- INL quantifies the overall deviation of the transfer curve from an ideal straight line.

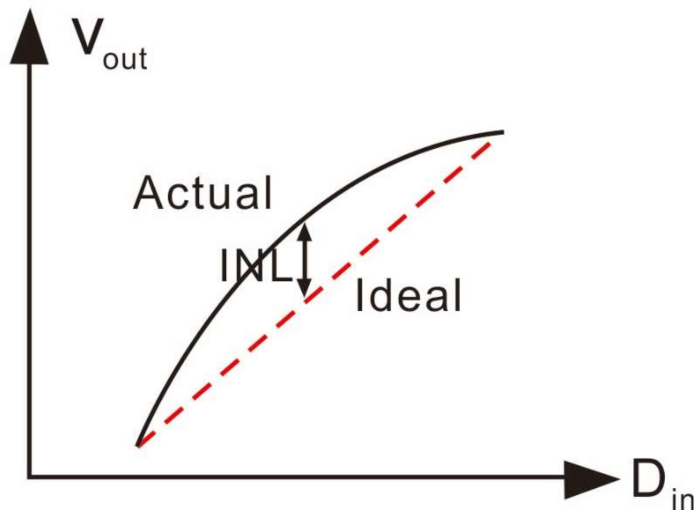


General Idea for INL

ADC



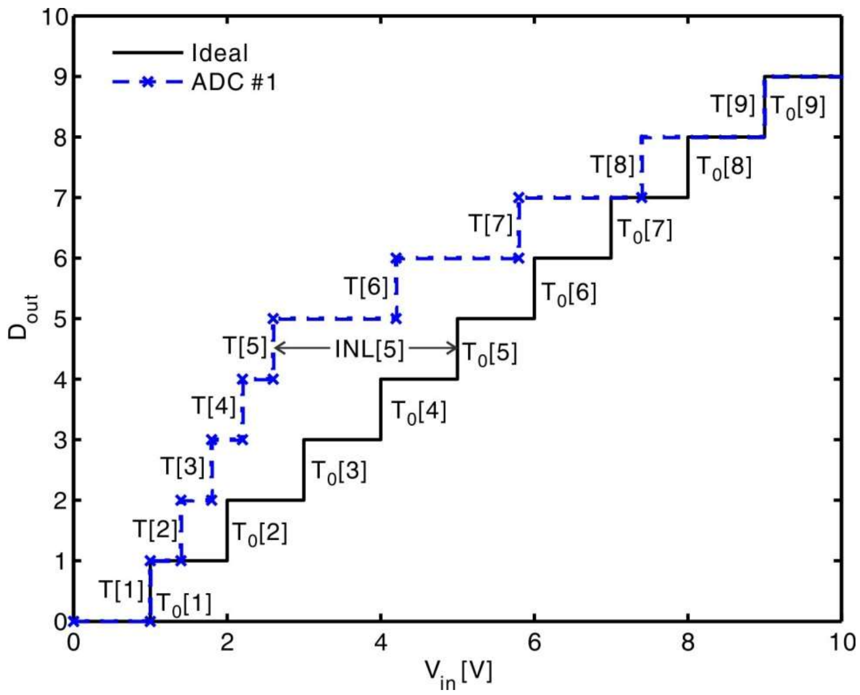
DAC



ADC INL Example

$$\mathrm{INL[k]} = \frac{\mathrm{T[k]} - \mathrm{T_0[k]}}{\mathrm{W_{avg}}}$$

- $T_0[k]$ is the ideal evenly spaced transition level
- INL(0) is undefined
- INL(1) = INL(9) = 0
- Similar as DNL, INL is also independent from gain and offset errors



Relationship between INL and DNL

- It is easy to derive that:

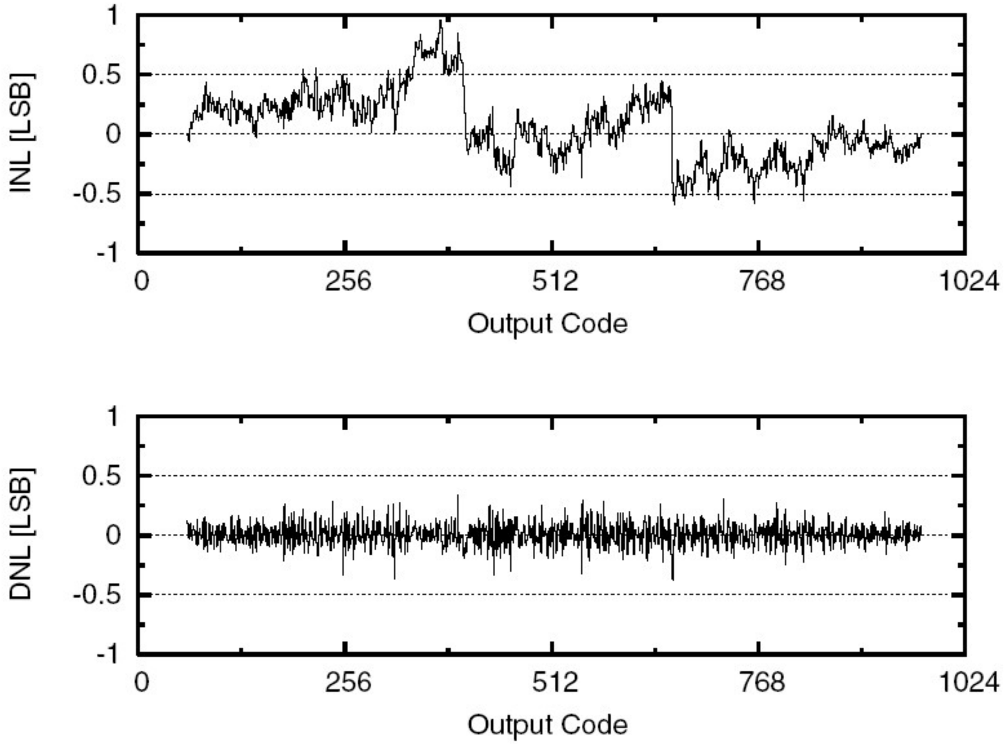
$$\mathrm{INL}[\mathrm{k}] - \mathrm{INL}[\mathrm{k}-1] = \mathrm{DNL}[\mathrm{k}-1] \quad \Rightarrow \quad \mathrm{INL}(\mathrm{k}) = \sum_{\mathrm{i}=1}^{\mathrm{k}-1} \mathrm{DNL}(\mathrm{i})$$

- Once we know DNL, we can easily find INL.

	ADC #1		ADC #2	
Code [k]	DNL [LSB]	INL [LSB]	DNL [LSB]	INL [LSB]
1	-0.6	0	-0.6	0
2	-0.6	-0.6	+0.6	-0.6
3	-0.6	-1.2	-0.6	0
4	-0.6	-1.8	+0.6	-0.6
5	+0.6	-2.4	-0.6	0
6	+0.6	-1.8	+0.6	-0.6
7	+0.6	-1.2	-0.6	0
8	+0.6	-0.6	+0.6	-0.6
9	Undefined	0	Undefined	0

- ADC #1 is more nonlinear.

Typical ADC DNL/INL Plot



[Ishii, CICC, 2005]

- DNL/INL plots reveals architectural details.
- A well-designed ADC has both INL and DNL bounded by $\pm$1 LSB

DAC INL

- Same idea applies
  - Find ideal output values that lie on a straight line between endpoints
  - Calculate INL for each code k using

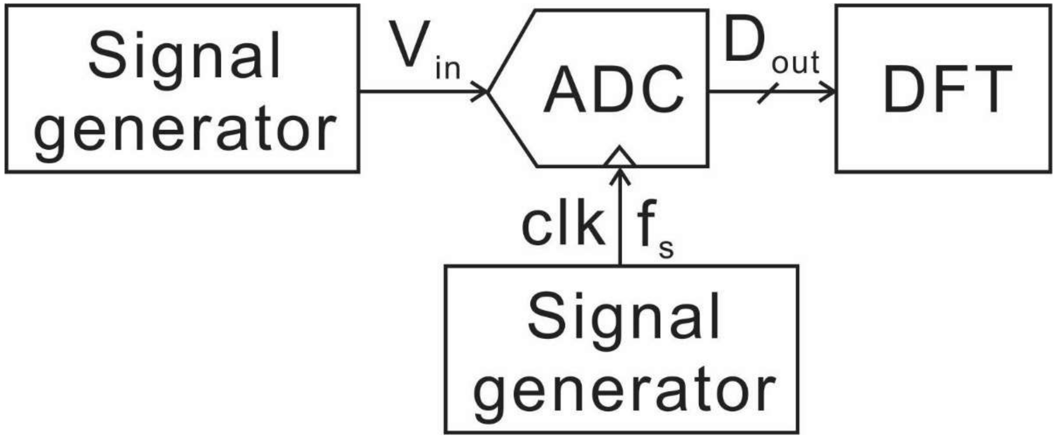
$$\mathrm{INL}[k] = \frac{V_{out}[k] - V_{out,0}[k]}{\mathrm{Step}_{avg}}$$

  - $V_{out,0}[k]$ is the ideal output voltage for code k

Spectral Metrics

- Static metrics cannot capture dynamic errors.
- ADC/DAC performance is a function of frequency.
- Effective performance evaluation in the frequency domain
  - Apply sine waves at the input and look at the output spectrum
  - Expect same sine tones at the output, all other frequency components represent nonidealities
- List of spectral metrics:
  - SNR: Signal-to-noise ratio
  - SNDR (SINAD): Signal-to-(noise + distortion) ratio
  - ENOB: Effective number of bits
  - DR: Dynamic range
  - SFDR: Spurious free dynamic range
  - THD: Total harmonic distortion
  - ERBW: Effective Resolution Bandwidth
  - IMD: Intermodulation distortion
  - MTPR: Multi-tone power ratio

ADC Spectral Metrics Test Setup



- What are the differences between DFT, CTFT, and DTFT?

Fourier Transforms (1)

- Continuous-time FT:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

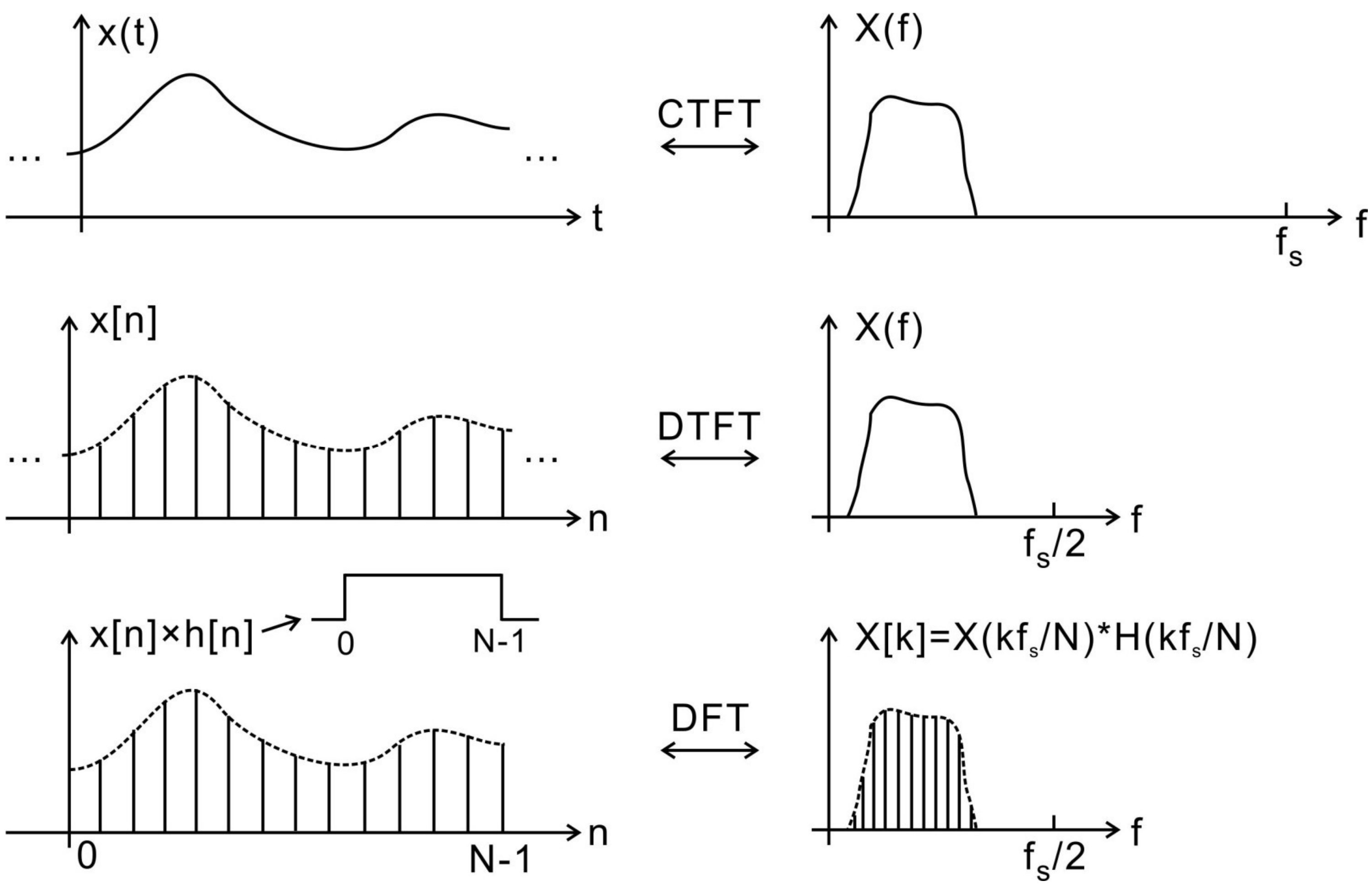
- Discrete-time FT (Z transform):

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j}$$

- Discrete FT:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi k \quad /N}$$

Fourier Transforms (2)



- h[n] is a rectangular window. Is there any distortion in the spectrum?

Discrete Fourier Transform

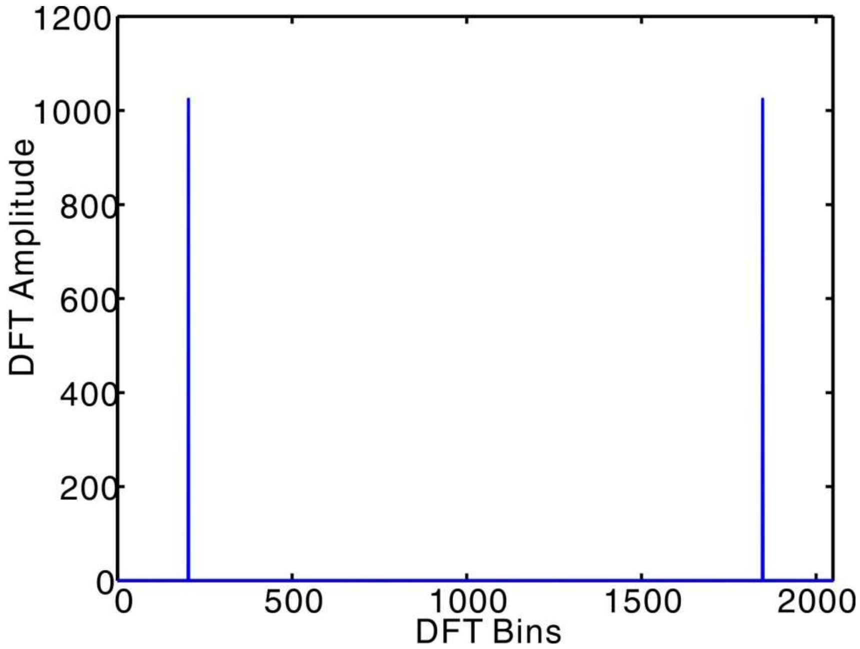
- DFT takes a block of N time-domain samples (spaced T_s=1/f_s) and produces a set of N frequency bins

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi k \quad /N}$$

- Bin k represents frequency content at $k \cdot f_s/N$
- DFT frequency resolution
  - Proportional to $f_s/N = 1/(N \cdot T_s)$
  - $N \cdot T_s$ is total time spent on gathering samples
- Parseval's theorem: $$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$$

Direct DFT Plot

- N = 2048; fs = 1024; fin = 101;
- Dout = Vin;
- Dout_fft = abs(fft(Dout));



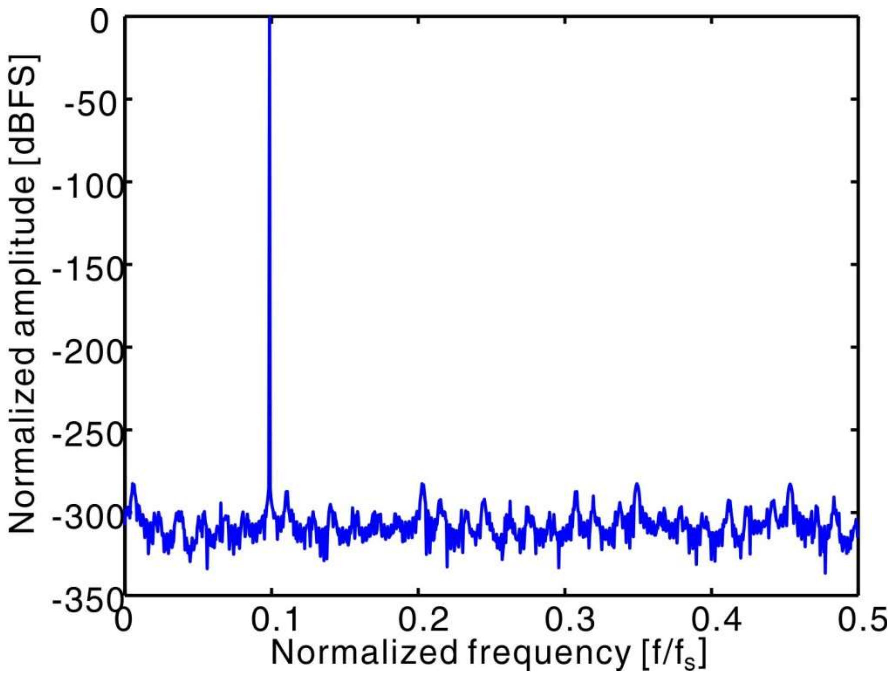
- Drawbacks
  - Amplitude depends on N
  - Do not know the frequency
  - Cannot clearly see the spectrum other than the fundamental tones
  - Have two identical half

Normalized Plot

Improvements:

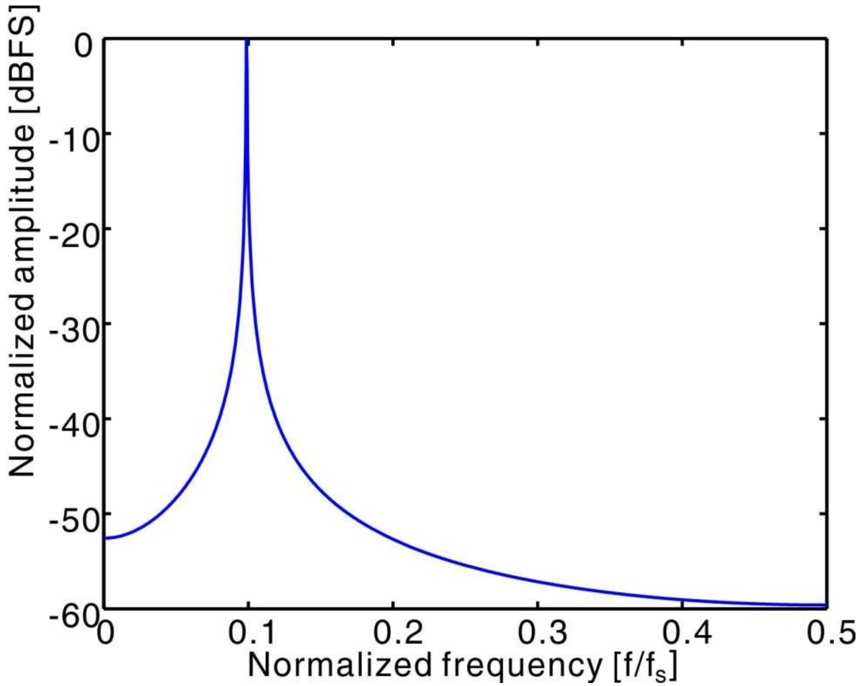
- Normalize the amplitude [dBFS]
- Normalize the frequency [f/f_s]
- Plot the amplitude in log scale
- Remove the redundant half

Question: why there is a noise floor?

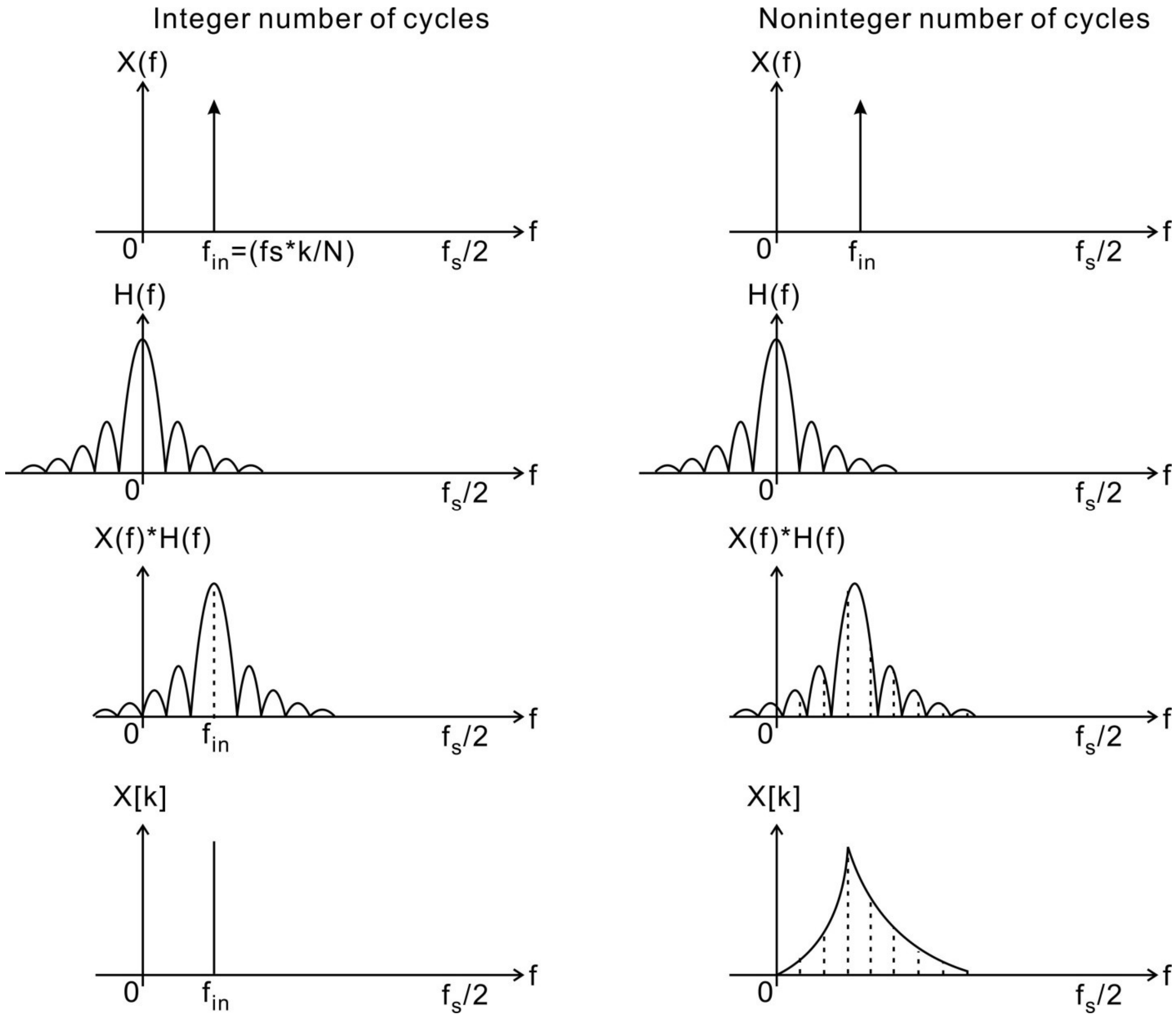


Change Frequency

- All things remain the same except fin = 101→101.2
- What's wrong?



Spectral Leakage

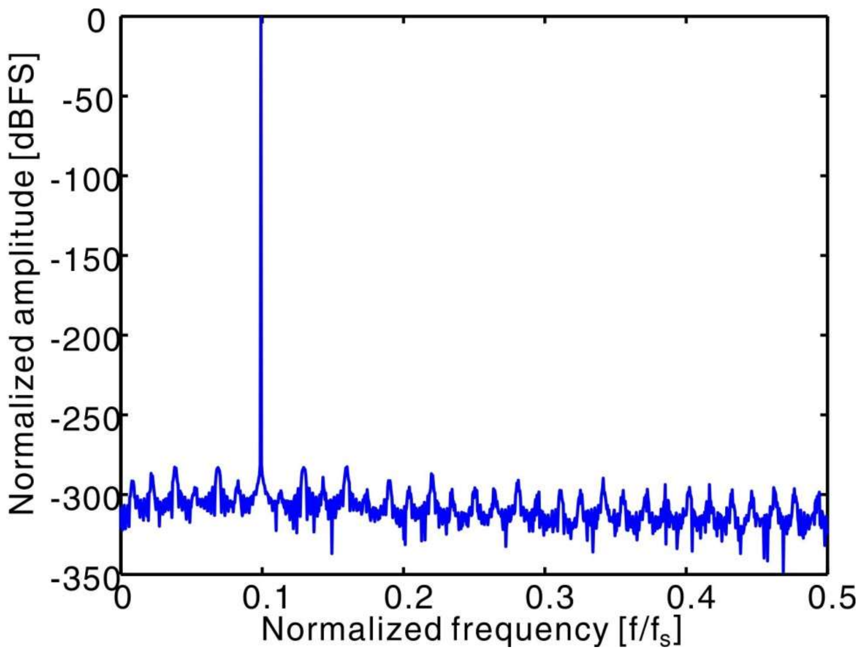


Summary

- DFT is the sampled spectrum of $x[n]\times h[n]$
  - h[n] is a rectangular window.
- If x[n] is not periodic of N, there is spectral leakage.
- 2 ways to solve this problem
  - Ensure integer number of cycles
  - Use other window functions with smaller side lobes

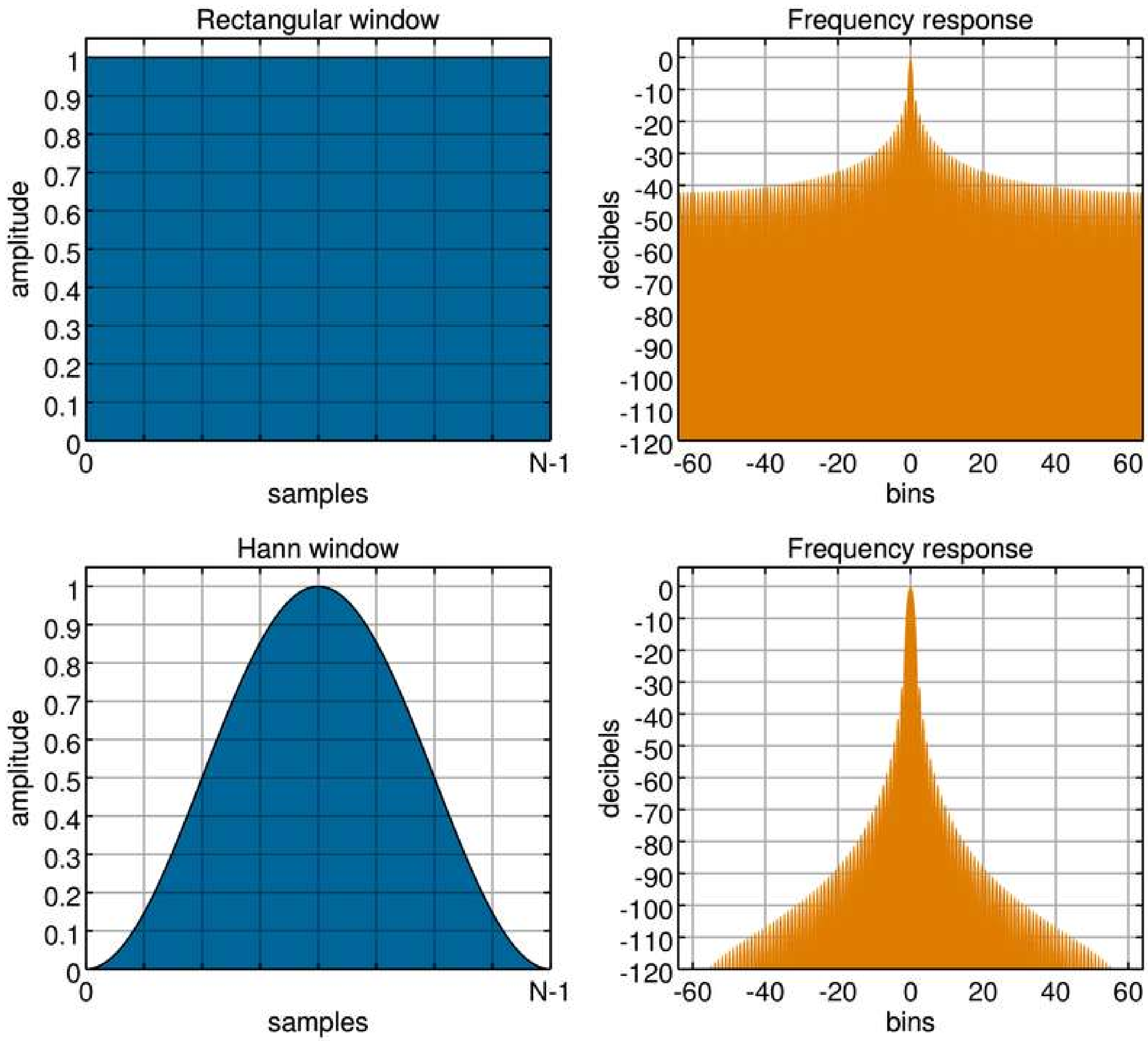
Ensure Integer Number of Cycles

- N = 2048; fs = 1024;
- fin = fs/N*cycle;
- cycle must be an integer
  - E.g., cycle = 203;
  - fin = fs/N*cycle = 101.5
- How many usable frequencies are there in (0, fs/2)?
  - 0 < cycle < N/2

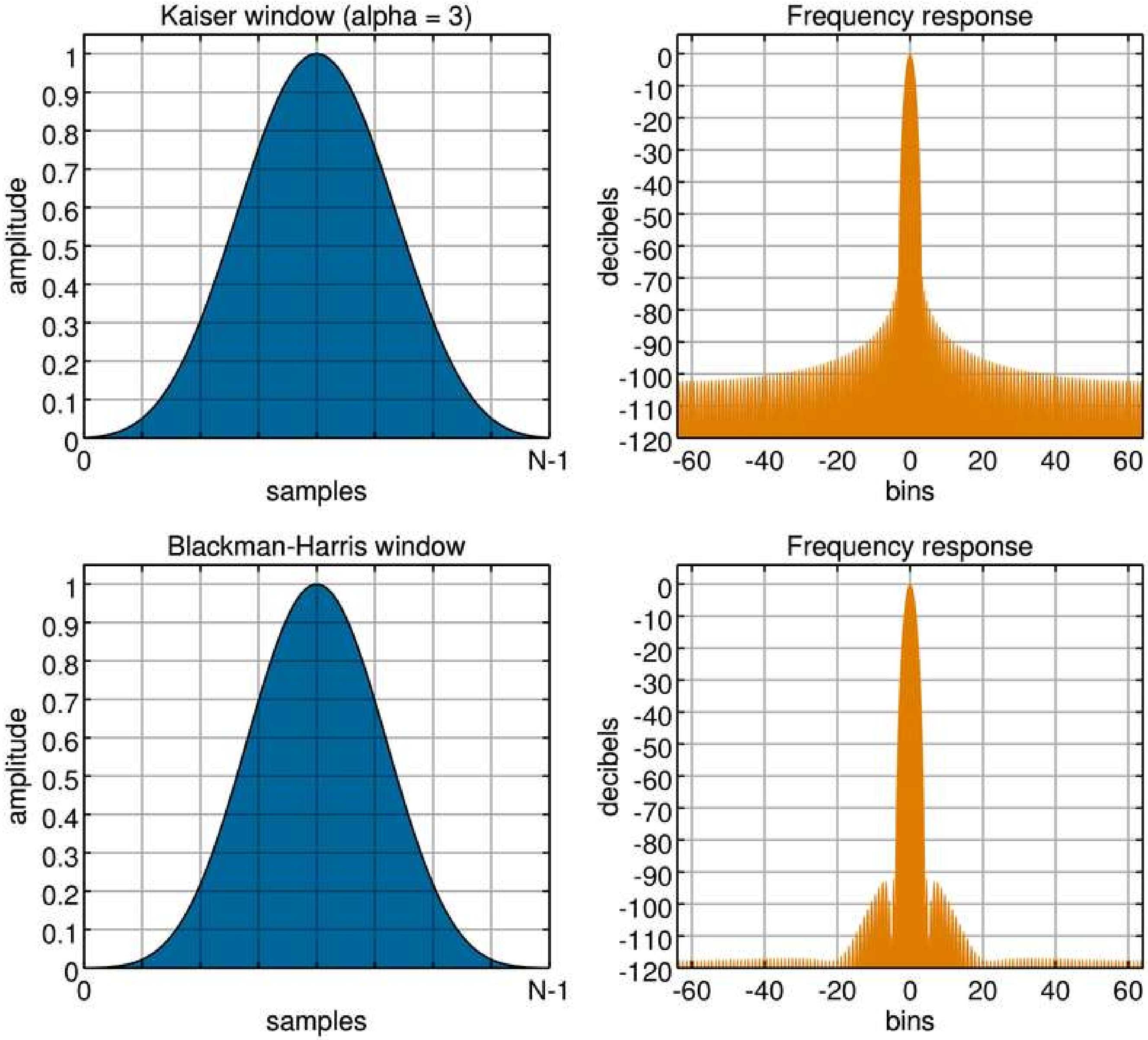


Windowing

- Use a different window function with smaller side lobes than that of the rectangular window
- Many window functions to choose from:
  - Hann, Hamming, Blackman, Kaiser, Cosine, Hodie, …
  - Tradeoff: main-lobe width vs. side-lobe attenuation
- The width of the main-lobe is now larger
  - The signal spans more-than-1 frequency bins

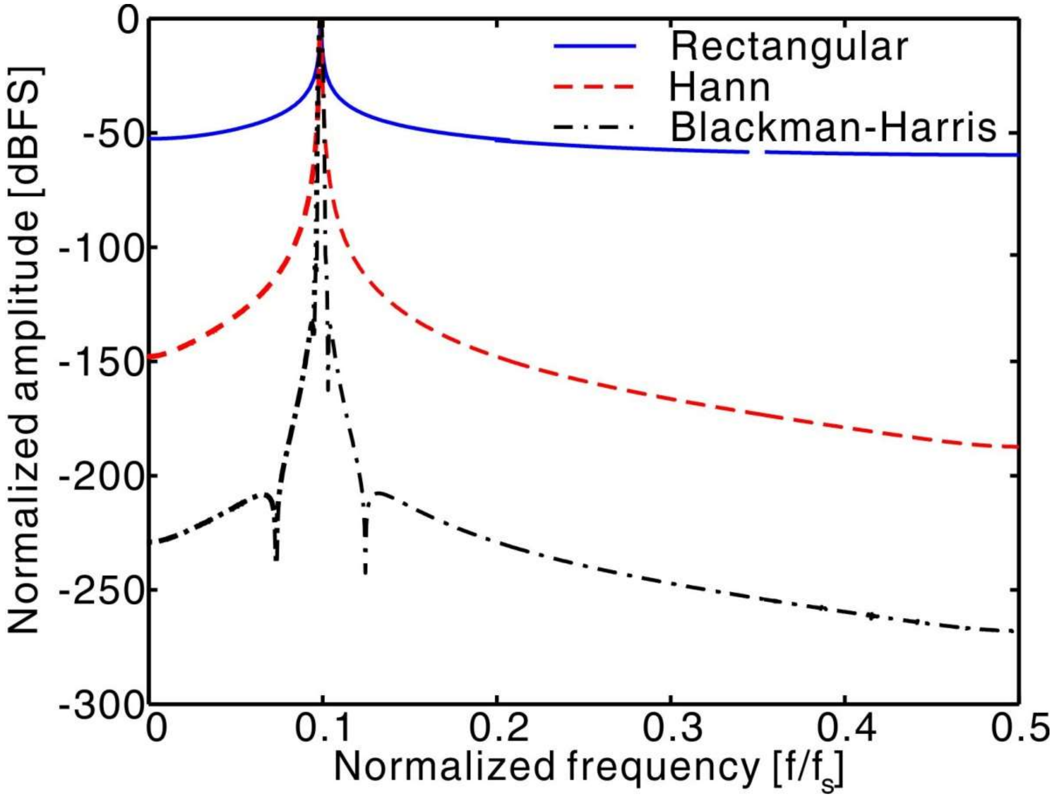


http://en.wikipedia.org/wiki/Window_function



http://en.wikipedia.org/wiki/Window_function

Spectrum with Windowing



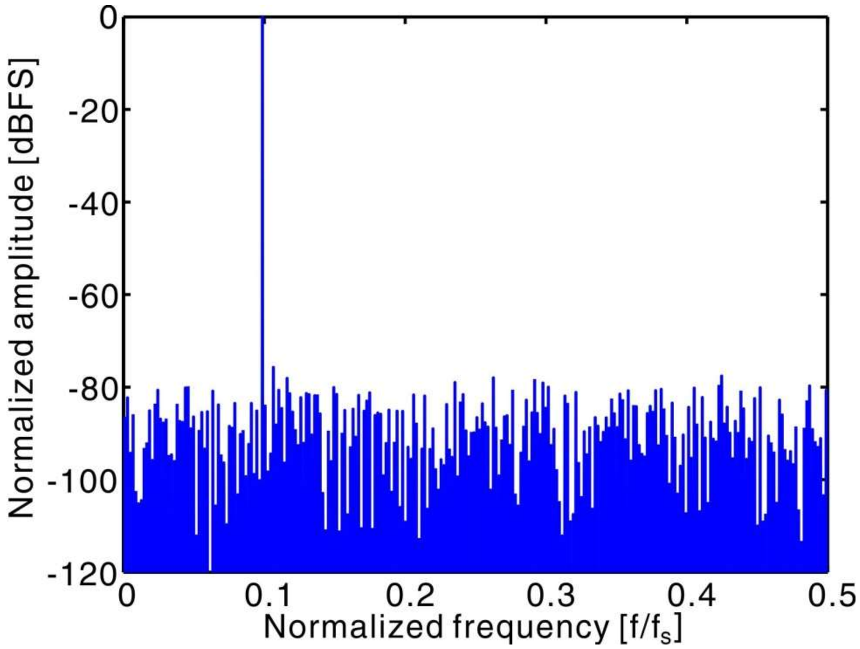
Dout = Dout.*window_func

Integer Cycles vs. Windowing

- Integer number of cycles
  - Input falls into a single DFT bin ☺
    - No smearing
    - Highest spectral resolution
  - Requires careful choice of the input frequency ☹
    - Ideal for simulations
    - In real measurements, we can set the input signal to be a fraction of the clock signal
      - “Coherent sampling"
- Windowing
  - No restrictions on signal frequency ☺
  - Signal and harmonics distributed over several DFT bins ☹
    - Be careful about the smeared out nonidealities
    - Requires more samples for a given accuracy

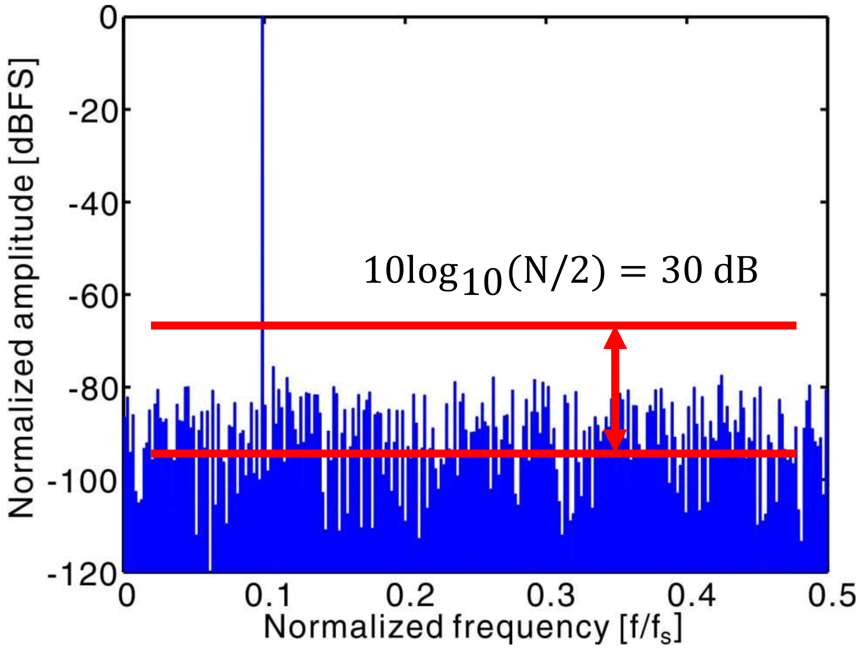
Spectrum with Quantization Noise

- N = 2048; fs = 1024; fin = 101;
- B = 10;
- LSB = 2/2^B;
- Dout = round(Vin/LSB)*LSB;
- Noise spectrum looks flat
- How to calculate SQNR from FFT?
  - Use Parseval's theorem
  - Time-domain power = frequency-domain power
  - SQNR = 6B+2 = 62 dB
- Why noise floor is not at -62 dBFS?



Noise Floor

- Total quantization noise is distributed in all fft bins
- The total noise is -62 dBFS, and thus, the noise floor (which is the noise in each bin) is -62 dBFS/(N/2) = -92 dBFS
- The noise floor depends on N, and thus, the total noise cannot be inferred if N is not specified.

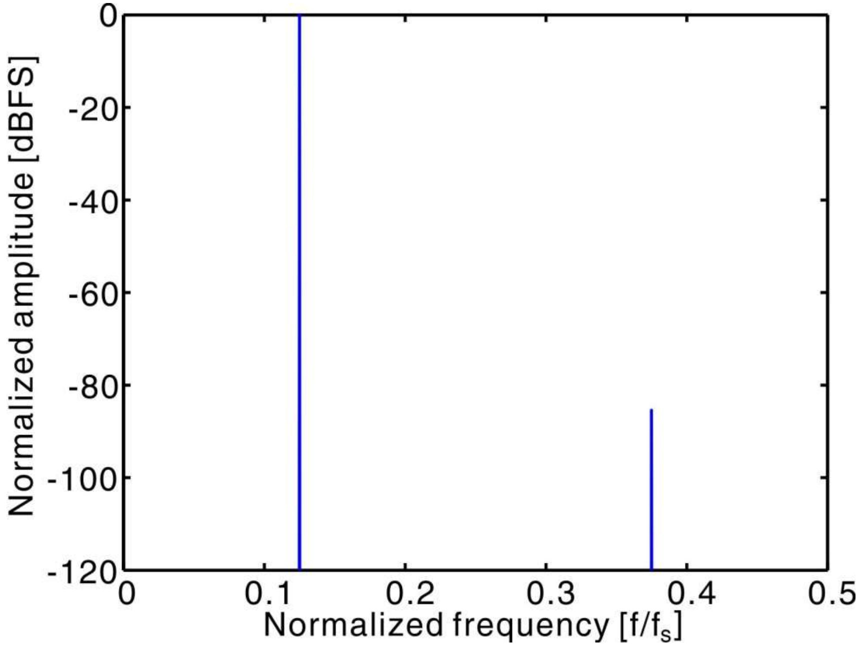


DFT Plot Annotation

- Most common annotation
  - Specify N: how many DFT points were used
- Less common options
  - Shift DFT noise floor by $10\log_{10}(N/2)$ dB
  - Normalize the bin width to 1 Hz and express noise as power spectral density (similar to continuous-time FT)
    - "Noise power in 1 Hz bandwidth"

Periodic Quantization Noise

- fin = 101 → 128
- fin = fs / 8
- Quantization noise is repetitive and highly correlated with the input.
- To make noise spectrum flat and span all bins, it is important to make N and cycle mutually prime
  - GCD(N,cycle) = 1



Typical ADC Output Spectrum

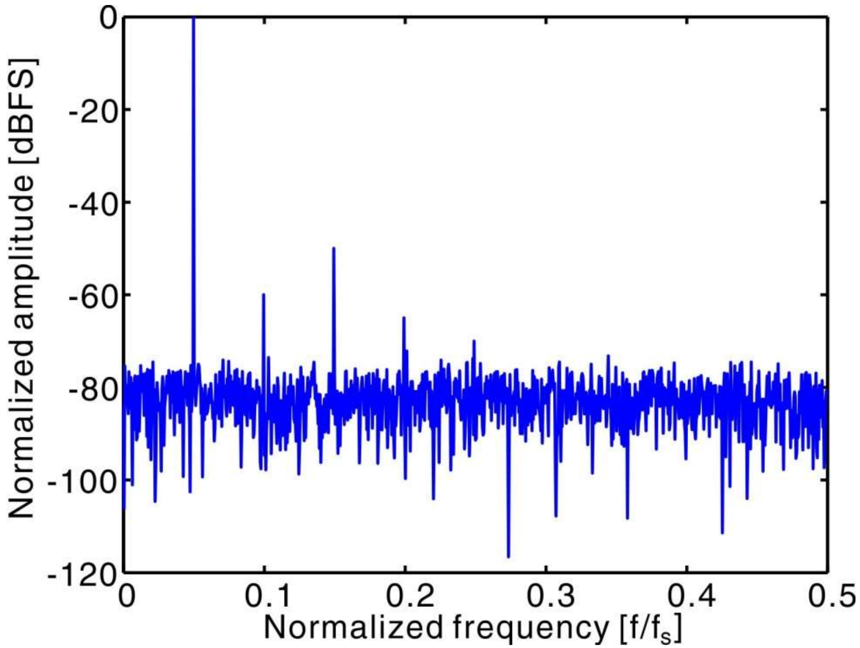
- Harmonics due to nonlinearities
- Signal-to-distortion ratio (SDR)

$$\text{SDR} = \frac{\text{Signal Power}}{\text{Total Distortion Power}}$$

- Total harmonic distortion

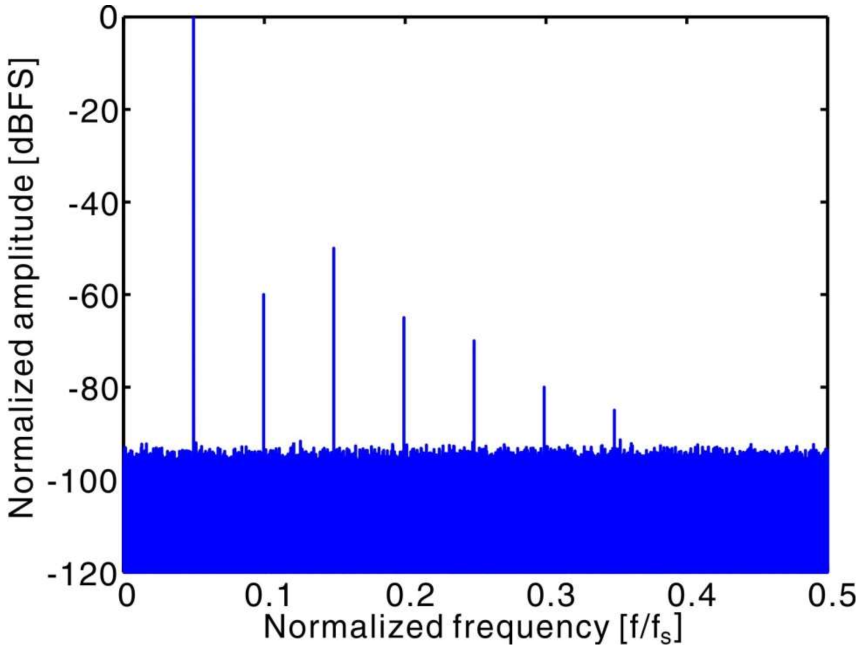
$$\text{THD} = \frac{1}{\text{SDR}}$$

- By convention, total distortion power consists of 2nd through 7th harmonics
  - Other distortion induced harmonics are ignored, why?
- Are there 6th and 7th harmonics?



Lowering the Noise Floor

- We can lower the noise floor by increasing the FFT size.
- $N = 2^{11} \rightarrow 2^{18}$
- We clearly see all harmonics.

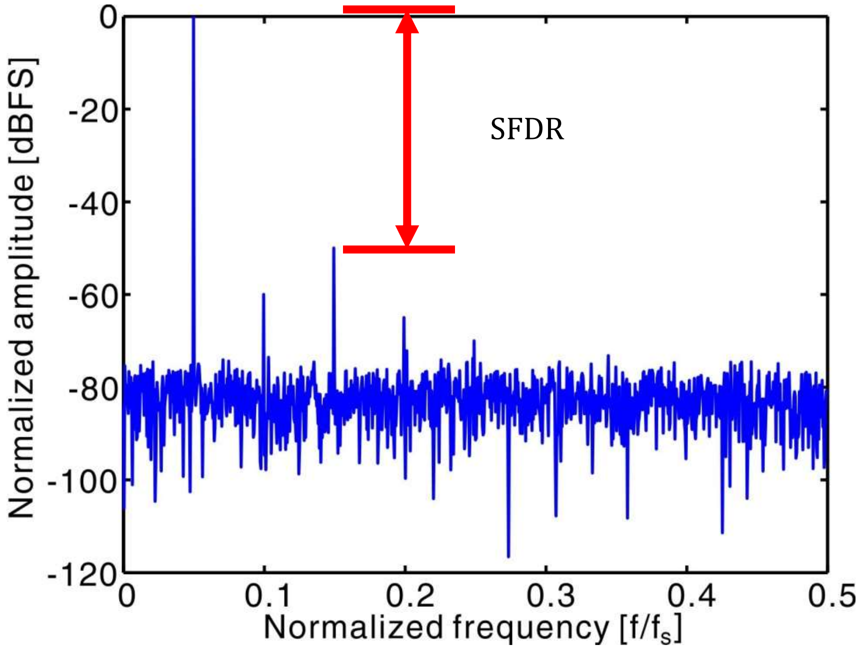


SFDR

- Spurious free dynamic range (SFDR)

$$\mathrm{SFDR} = \frac{\text{Signal Power}}{\text{Largest Spurious Power}}$$

- Largest spur is often, but not necessarily, an input harmonic
- Need to specify the input signal power, why?

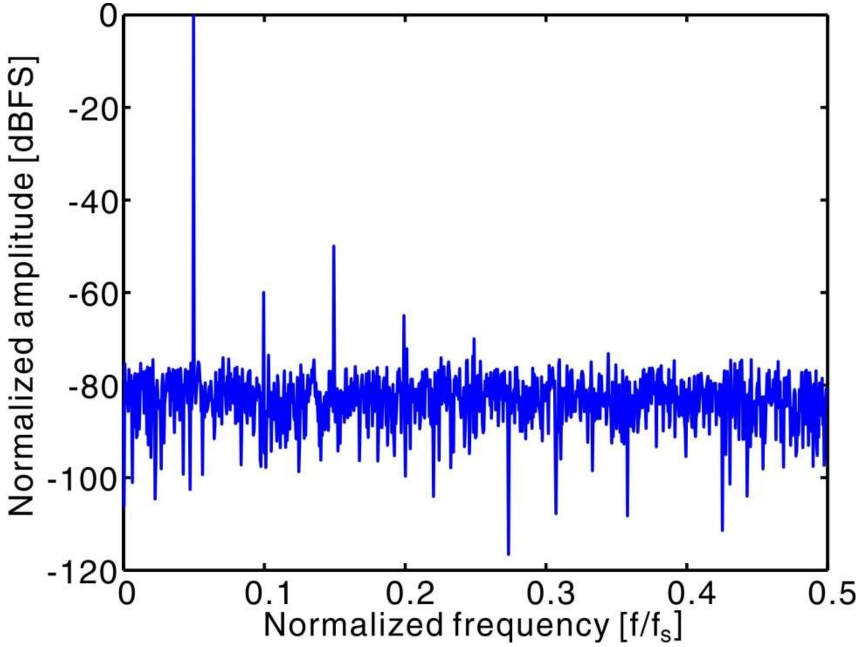


SNR

- Signal-to-noise ratio (SNR):

$$\mathrm{SNR} = \frac{\text{Signal Power}}{\text{Total Noise Power}}$$

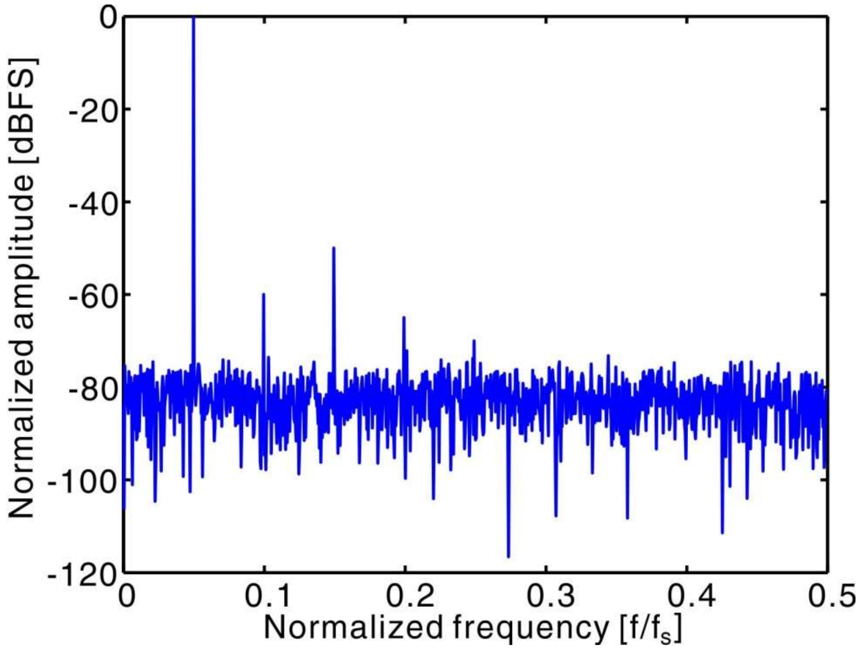
- Total noise power includes all bins except DC, input, and 2nd through 7th harmonic.
- It consists of:
  - Quantization noise
  - Electronic noise
  - “DNL noise”
  - Other distortion terms
    - Sampling errors



SNDR and ENOB

- Signal-to-(noise + distortion) ratio (SNDR):

$$\mathrm{SNDR} = \frac{\text{Signal Power}}{\text{Noise and Distortion Power}}$$



- Total noise and distortion includes all bins except DC and the input.

- Effective number of bit (ENOB):

$$\mathrm{ENOB} = \frac{\mathrm{SNDR\ (dB)} - 1.76}{6.02} < \mathrm{B}$$

Effective Number of Bits

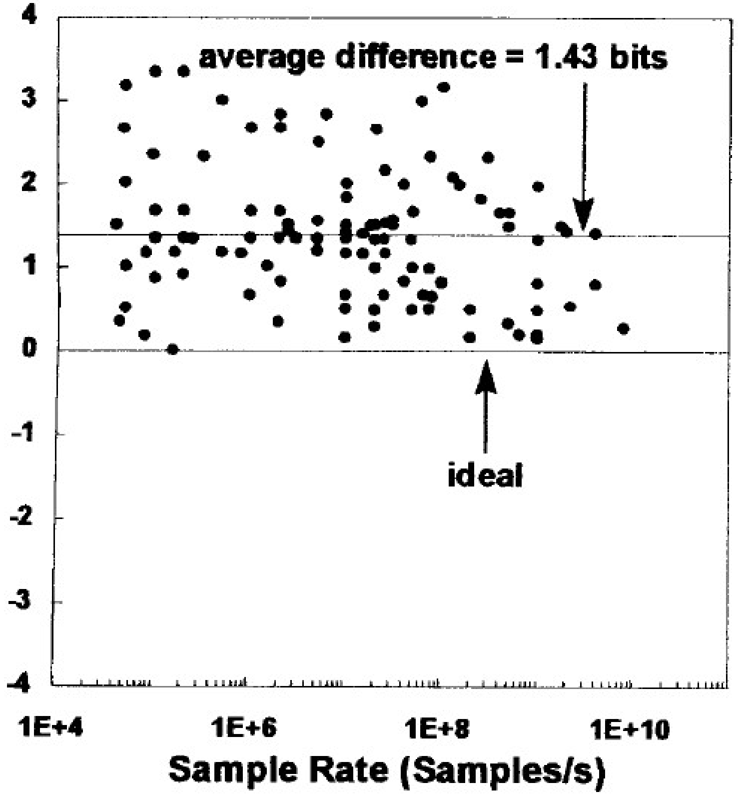
- What is the ENOB for a 10-bit converter with 50 dB SNDR?

$$\text{ENOB} \cong \frac{50-2}{6} = 8$$

- Assuming the stated resolution of an ADC is B, ENOB = B only for "0" electronic noise and "0" distortion
- Distortion can be lowered through better design.
  - For a well designed ADC, SNDR should not be limited by distortion.
- Electronic noise is more fundamental. It directly trades with power.
- Usually electronic noise is made comparable with the quantization noise.
  - Diminishing return for further reducing electronic noise
- For good power efficiency: ENOB < B-1

Survey Data

$\mathrm{B} - \mathrm{ENOB}$

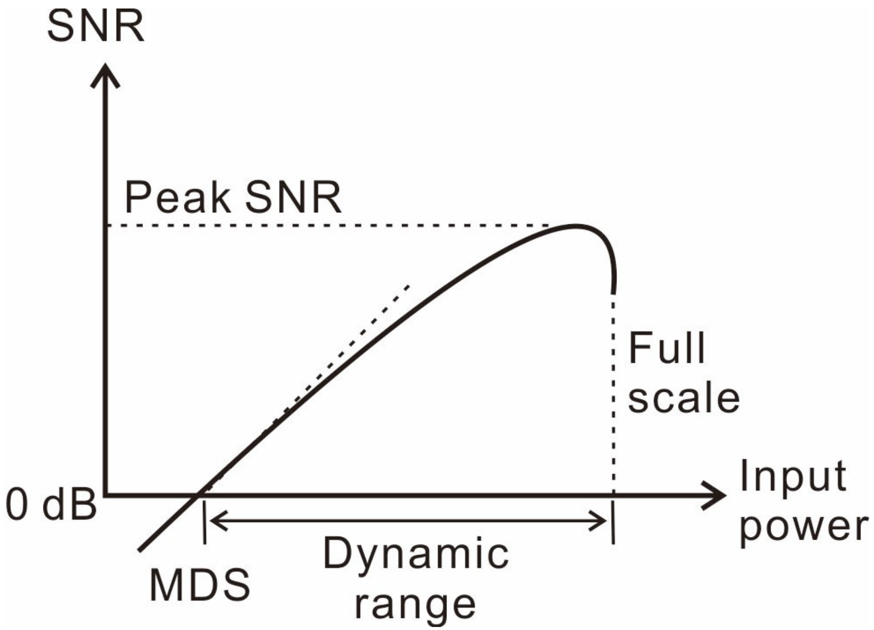


R. H. Walden, "Analog-to-digital converter survey and analysis," *IEEE J. on Selected Areas in Communications*, pp. 539-50, April 1999

Dynamic Range

- Dynamic range:

$$\mathrm{DR} = \frac{\text{Maximum Signal Power}}{\text{Minimum Detectable Signal Power}}$$



- Minimum detectable signal (MDS) is defined as the power of the input that produces 0 dB SNR.

- DR ≥ peak SNR
  - As signal amplitude increases, the noise floor may rise.

Aliasing

- Due to aliasing, harmonics can appear at frequencies that do not look like harmonics.

- For example:

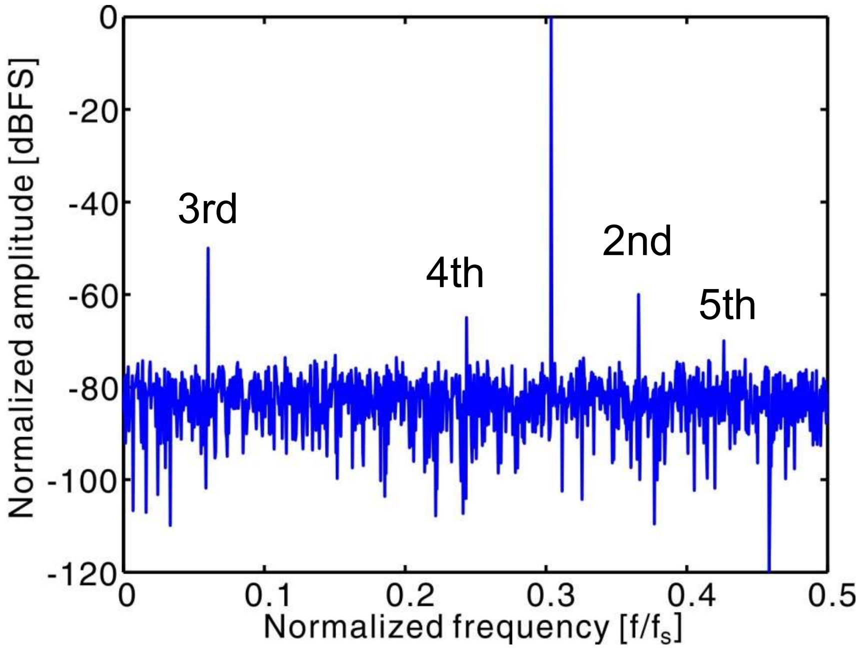
  $f_1 = f_{in} = 0.3125\ f_s$

  $f_2 = 2f_1 = 0.6250\ f_s \rightarrow 0.3750\ f_s$

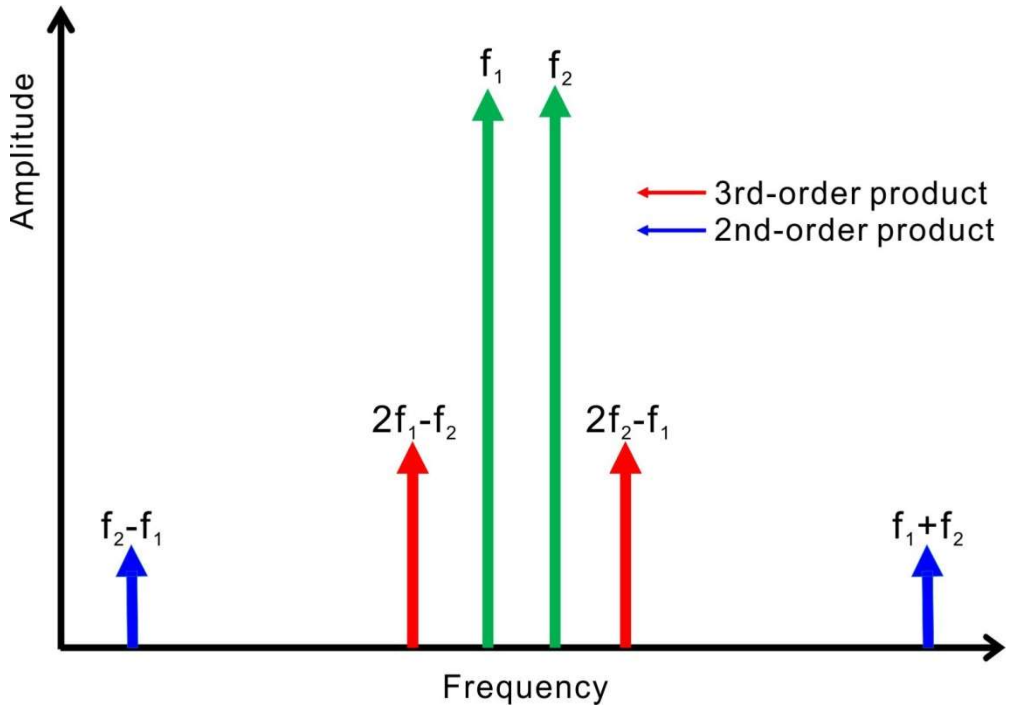
  $f_3 = 3f_1 = 0.9375\ f_s \rightarrow 0.0625\ f_s$

  $f_4 = 4f_1 = 1.2500\ f_s \rightarrow 0.2500\ f_s$

  $f_5 = 5f_1 = 1.5625\ f_s \rightarrow 0.4375\ f_s$

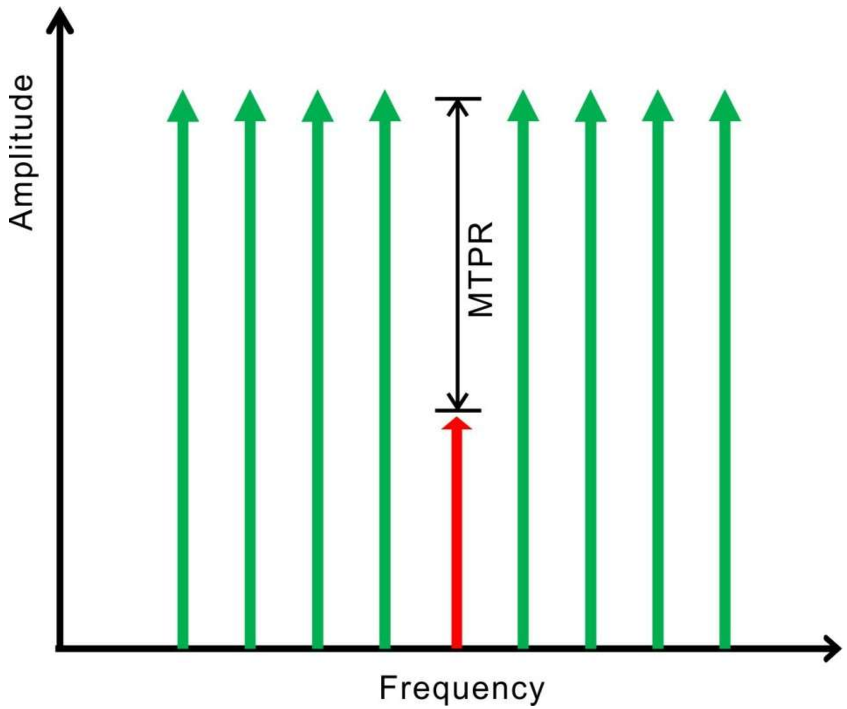


Intermodulation Distortion



- IMD is important in multi-channel communication
  - 3rd-order products are difficult to filter out

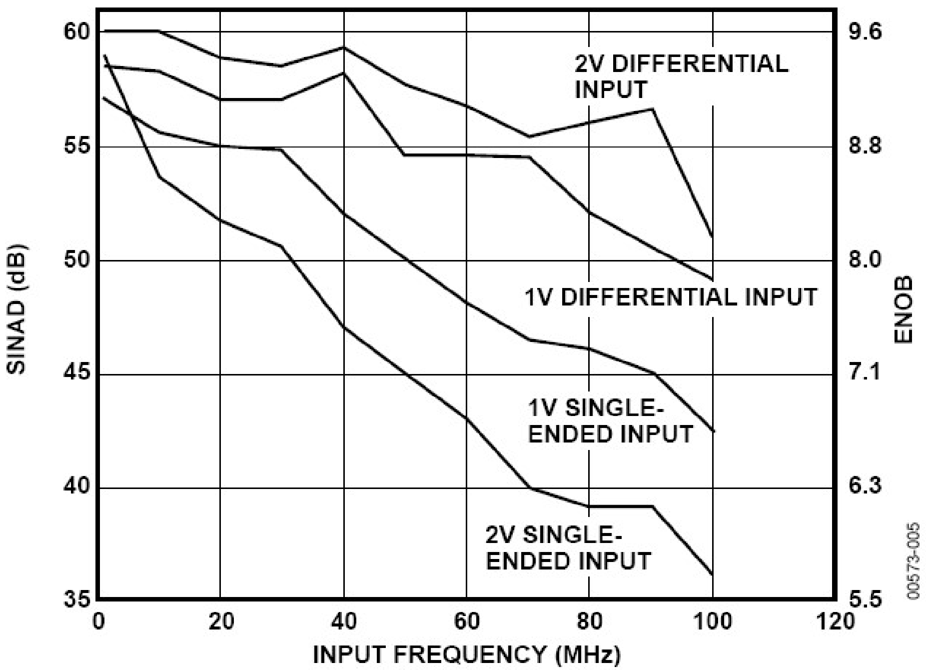
Multi Tone Power Ratio (MTPR)

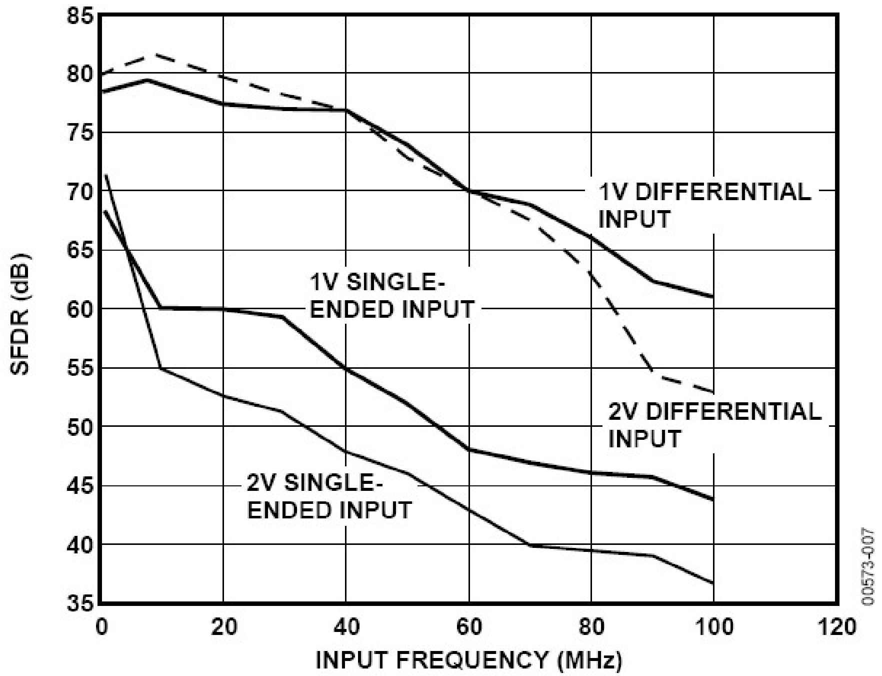


- MTPR is a useful metric in multi-tone transmission systems
  - E.g. OFDM

Frequency Dependence (1)

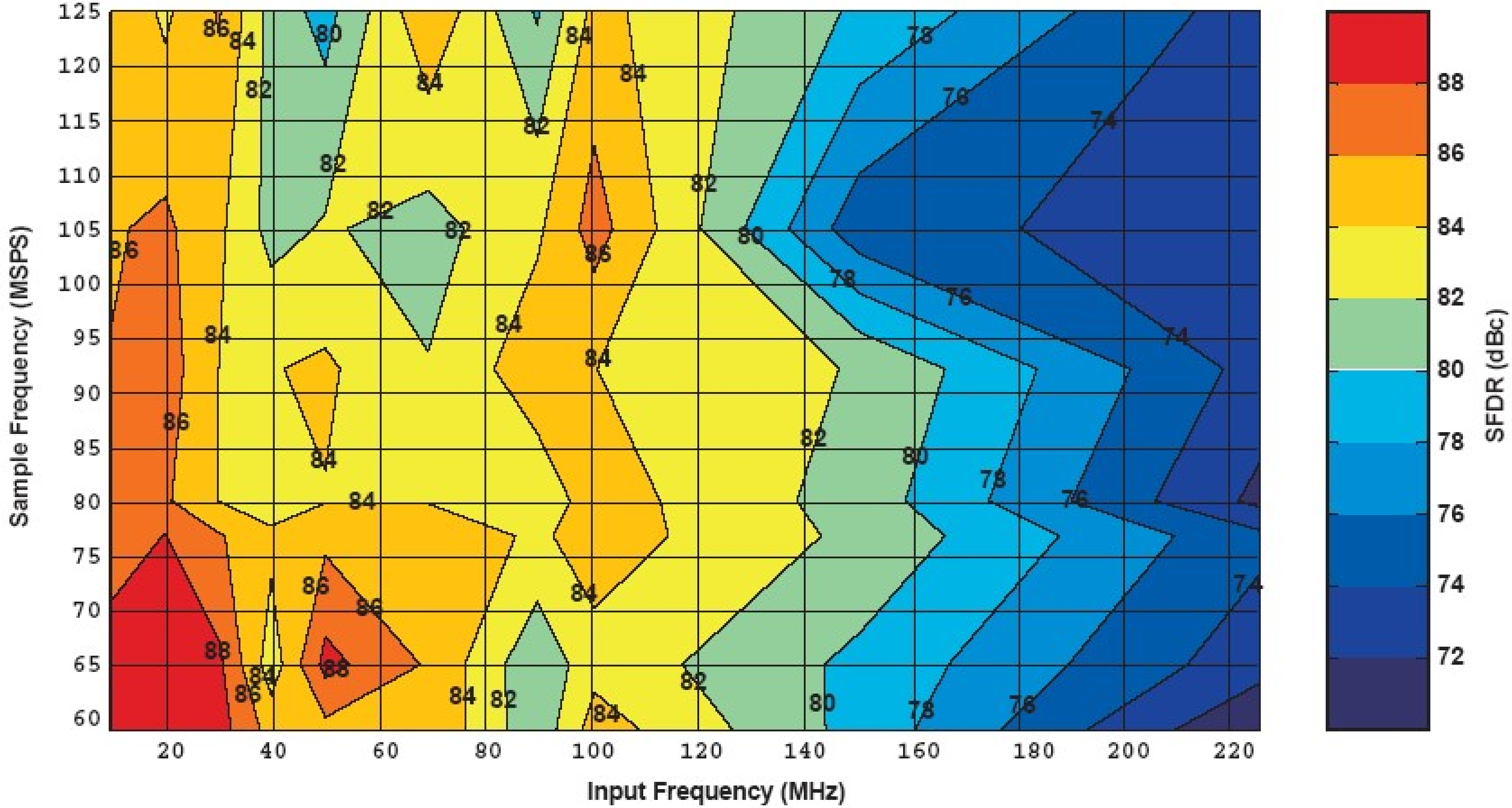
- Data converter performance depends on both the sampling and the input frequency





[Analog Devices, AD9203 Datasheet]

Frequency Dependence (2)

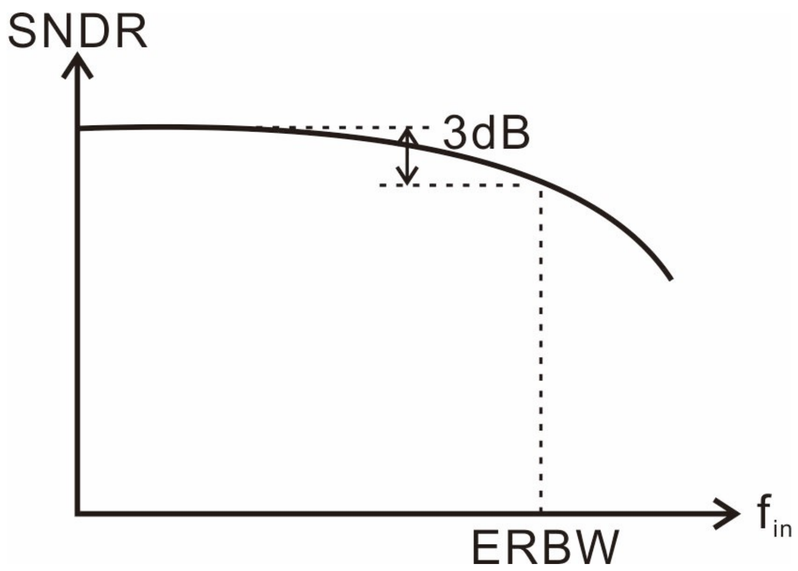


[Texas Instruments, ADS5541 Datasheet]

- An island surrounded by ocean

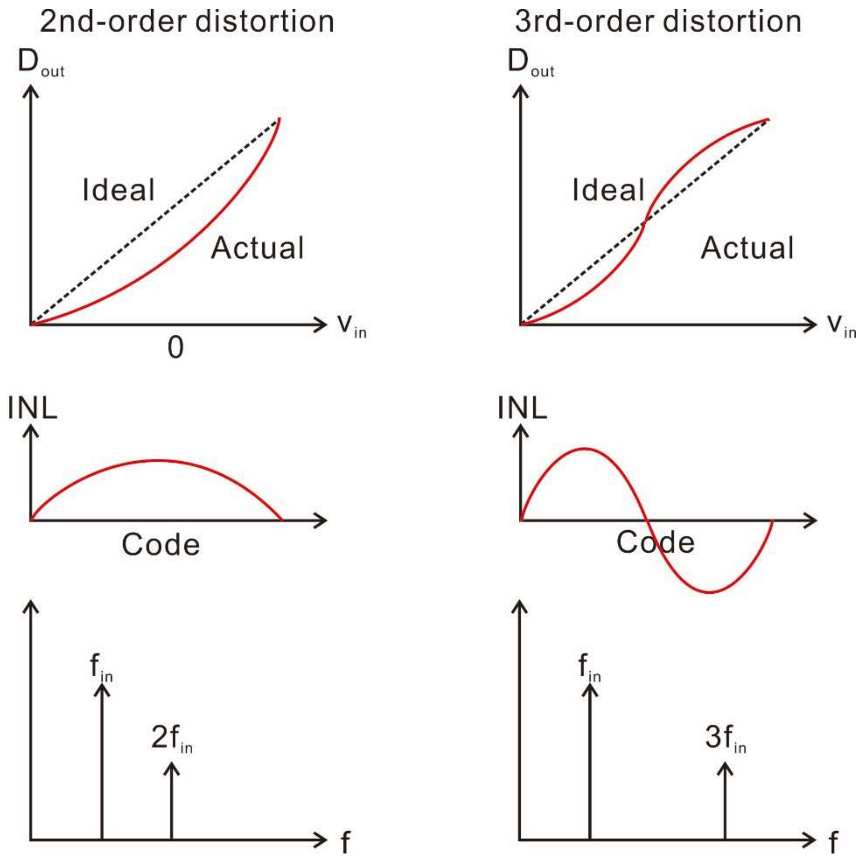
Effective Resolution Bandwidth (ERBW)

- ERBW is the input frequency at which the SNDR drops by 3dB compared to low frequency
  - 0.5-bit loss in ENOB
- ERBW $\approx f_s/2$ for a well designed Nyquist rate ADC
- ERBW > $f_s/2$ is not uncommon, especially for subsampling ADCs



INL/DNL and SFDR/THD/SNR

- Static nonlinearities (INL/DNL) and spectral nonlinearities (SFDR /THD/SNR) are closely related especially at low frequencies
  - At high frequencies, spectral nonlinearities tend to be limited by dynamic errors
- Can link systematic INL patterns with harmonics



SNR Degradation due to DNL (1)

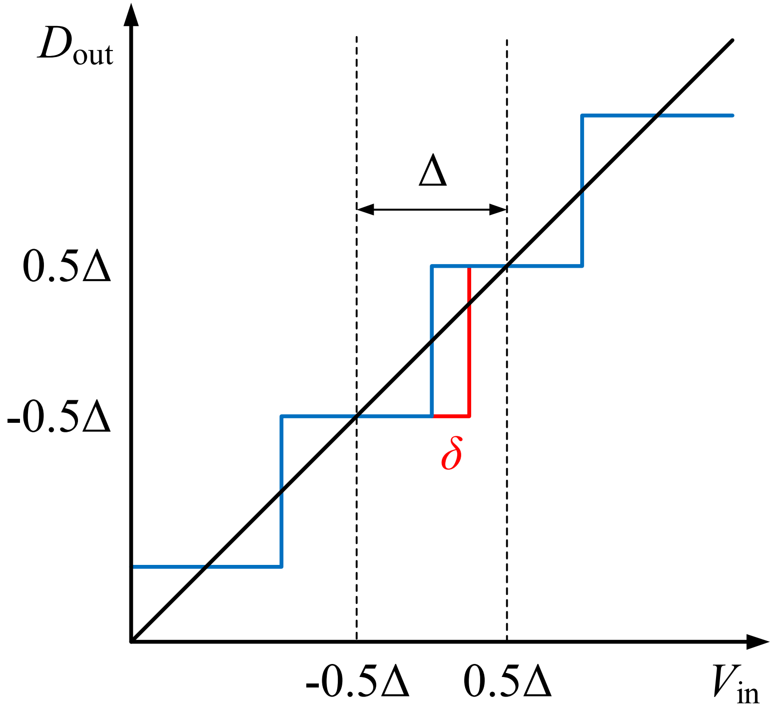
- If INL and DNL show irregular patterns, the nonlinearities may show up as “noise” like, which causes SNR degradation.
- Changes in the transition levels cause quantization noise to increase.
- For simplicity, let us consider a 1-bit quantizer with range of [-1,+1]. The transition level is changed from 0 to δ. As a result, the quantization noise is increased to:

$$\sigma^2 = \frac{1}{2}\left(\int_{\delta}^{1} (\mathrm{V_{in}} - 0.5)^2 \mathrm{dV_{in}} + \int_{-1}^{\delta} (\mathrm{V_{in}} + 0.5)^2 \mathrm{dV_{in}}\right) = \frac{1}{12} + \frac{1}{2}\delta^2$$

- Thus, if $\delta = 0.5$, the quantization noise is more than doubled.

SNR Degradation due to DNL (2)

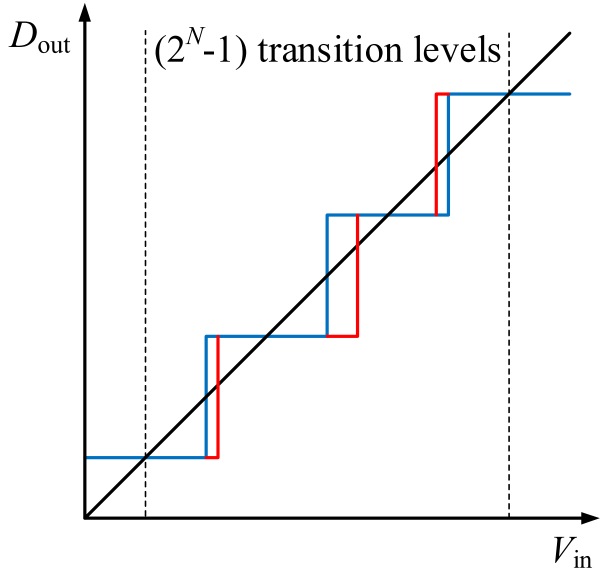
- For multibit quantization, assuming transition-level deviation δ is independent and uniformly distributed in [-0.5,+0.5] LSB
- This also translates to a DNL with triangular pdf distribution in [-1, +1]
- Assuming the input signal is uniformly distributed, the quantization error in [-0.5,+0.5]Δ is



$$\sigma_0^2 = \frac{1}{2^N\Delta}\left(\int_{-0.5\Delta}^{\delta}(V_{in}+0.5\Delta)^2 \mathrm{d}V_{in} + \int_{\delta}^{0.5\Delta}(V_{in}-0.5\Delta)^2 \mathrm{d}V_{in}\right) = \frac{1}{2^N}\left(\frac{\Delta^2}{12}+\delta^2\right)$$

SNR Degradation due to DNL (3)

- The amount of quantization noise increase introduced by one transition level deviation δ is $\delta^2/2^N$,and there are $(2^N\text{-}1)$ transition levels in total.

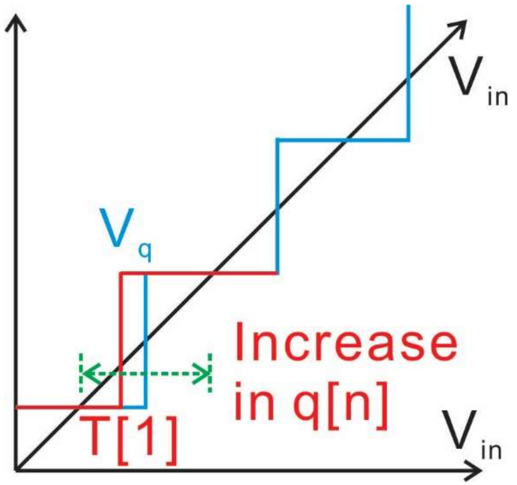


- Assuming that transition-level deviation δ is independent, the amount of quantization noise is

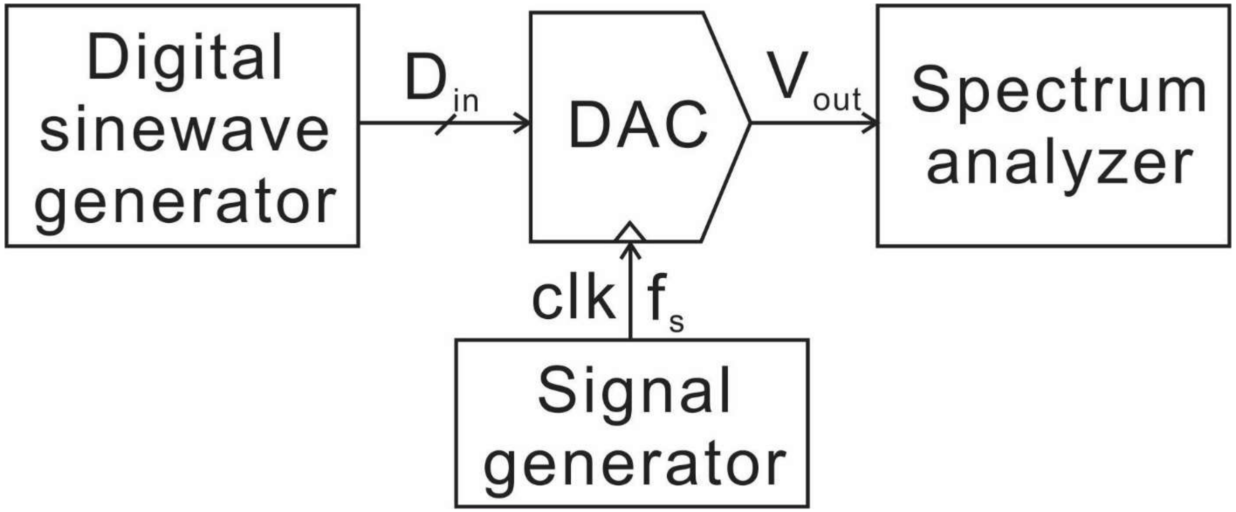
$$\sigma^2 = \frac{\Delta^2}{12} + \frac{2^N - 1}{2^N} E\left(\delta^2\right) \approx \frac{\Delta^2}{12} + E\left(\delta^2\right)$$

Transition-level, INL, and DNL noise

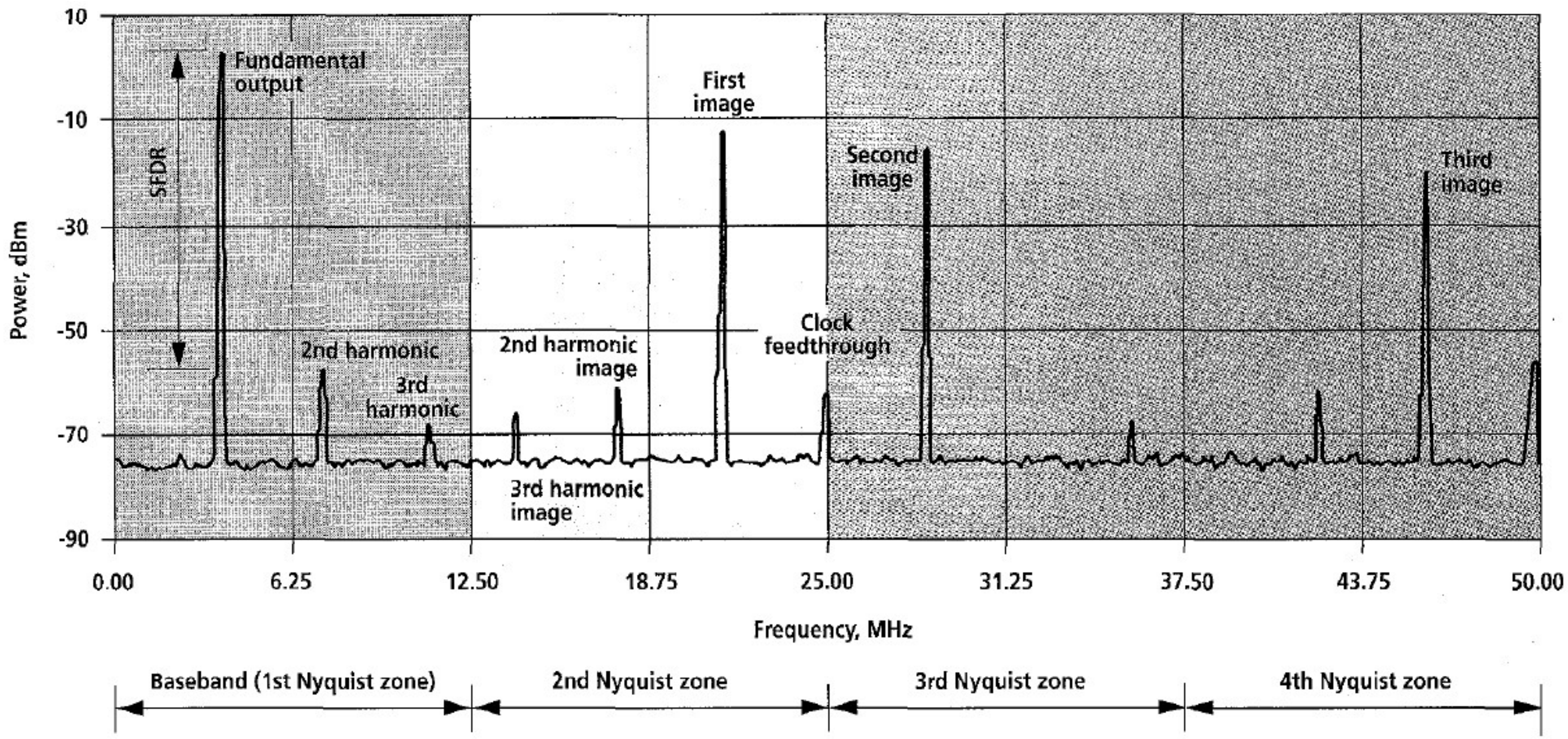
- Transition levels and INL are directly linked
- If we assume certain distribution of transition levels, it directly determines the distribution of INL.
  - For example, if we assume that transition level deviation δ follow independent and uniform distribution of [-0.5,+0.5] LSB, it directly means that INL follows the same distribution, because INL is by definition the transition level deviations.
  - DNL[i] = T[i+1]-T[i]-1 = δ[i+1] – δ[i]
    - Thus, DNL follows triangular distribution from -1 to 1 LSB.



DAC Test Setup



Typical DAC Output Spectrum



[Hendriks, "Specifying Communications DACs, IEEE Spectrum, July 1997]